Supplement to NeST-Com: Fixed-Budget Next-Scale Token Communication

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Implementation recipe, complete results, paired statistics and source examples. Numerical tables use the same frozen CSV sources as the main manuscript.

S1 Reproduction configuration

NeST-Com transmits source-token prefixes from a shared, frozen next-scale representation. The historical frame layouts, model manifest and reference vectors are supplied in `inputs/reproduction/`. The repository entry `experiments/nestcom-v9-completion-20261101/verify_cdf_ac.py` replays the integer coding checks on CPU; `verify_assets.py` resolves an external asset directory and checks file identities. No private server path is required for those verification commands.

S1.1 Representation, models and numerical route

The tokenizer has a 4096-entry codebook and ten maps of side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$. The VAR-d16 prior uses null class 1000 and greedy prediction, without CFG. Complete coarse maps determine every position's probability in the next map. Both terminals use those known maps; no unreceived same-scale token enters the receiver's entropy context.

The adapted decoder checkpoint is update0038000. It minimizes MSE+0.1 LPIPS on 20 000 ImageNet training images with a 50/25/25 mixture of continuous latents, fully quantized latents, and interpolated prefix-completed latents. The last input is $F_b + \alpha(F - F_b)$, $\alpha \sim U[0, 1]$, with F_b completed from eight source scales. Selection minimizes full-calibration mixture image loss. Tokenizer, prior and codebook remain frozen. The recipe is `experiments/var-latent-enhancement-20260917/configs/experiment.json`.

The NeST reference route is Python 3.10.12, NumPy 2.2.6, Torch 2.11.0+cu128, CUDA 12.8 and cuDNN 91900. Inference is FP32, batch 1, eval/no-grad, TF32 off, matmul precision highest, deterministic algorithms and cuDNN deterministic on, benchmark off. The revised timing uses two intra-op and two inter-op threads on two distinct physical CPU cores; S7 separates it from the historical SMT-pair run. Device PCI/UUID, tensor-state hashes, checkpoint-file hashes and source-file hashes are separate manifest fields. Reproducing a floating-point probability stream requires this numerical route as well as the weights.

Ada retains LlamaGen-L/VQ16 (upstream commit `ce98ec41803a74a90ce68c40ababa9eaeffeb4ec`), predicted ConvNeXt-tiny classes, conditional-branch entropy CDFs and CFG 4/T1/top-k2000/top-p1 sampled completion. Its 10-bit class is paid. The public LlamaGen weight URLs and revision are included in the manifest. ARPC uses its public 0.06M preset, CFG 3, temperature 0.5 and seed 0; its paid caption and source container are preserved. Latent JSCC uses MSE+0.1 LPIPS+0.01 latent consistency and shared D_c, with 100k updates; SwinJSCC uses RGB MSE, 200k updates and minimum calibration-MSE selection. Both use the fixed 20k training subset. External models retain their own numerical routes rather than inheriting NeST's environment.

Public VAR/LlamaGen/ConvNeXt/ARPC sources, exact file pins, available tensor-state pins and architecture configurations are listed in `model_manifest.json`. Project-trained D_c and JSCC checkpoints are retained in the project storage; no public download URL is asserted for them. Their recipes and exact checkpoint identities are supplied. Set `NESTCOM_ASSET_ROOT`, or provide `NESTCOM_ASSETS_JSON` mapping checkpoint names to local paths. A file SHA is never presented as a tensor-state SHA.

S1.2 Integer probabilities and independently terminated streams

FP32 logits pass through `torch.log_softmax`; NumPy float64 then performs stable normalization. For vocabulary $V = 4096$ and total frequency $M = 2^{24}$,

$$p_j = \frac{\exp(\ell_j - \max_k \ell_k)}{\sum_k \exp(\ell_k - \max_k \ell_k)}, \qquad f_j = \lfloor p_j(M - V) \rfloor + 1,$$

$$d = M - \sum_j f_j, \qquad f_{j^*} \leftarrow f_{j^*} + d, \quad j^* = \arg\max_j p_j.$$

Exact ties use the smallest codebook index. The int64 CDF is $[0, \mathrm{cumsum}(f)]$, with strictly positive frequencies. No CDF table is transmitted. The implementation is `src/var_comm/entropy.py`.

The arithmetic interval starts at $[0, 2^{32} - 1]$. An incremental encoder evaluates successive prefixes; each is finalized independently, without changing the incremental state. Finalization emits $b = 0$ if low$< 2^{30}$, otherwise $b = 1$, followed by pending+1 inverse bits. RX initializes a 32-bit state, uses bounded zero extension, and requires exactly 30 terminal zero reads. It re-encodes recovered tokens and requires exact equality to the received payload. The paid (m, K) specifies the token count. Prefix streams are not slices of a longer codeword. Missing files and invalid model/CDF inputs are execution errors, not receiver successes or NULLs.

S1.3 Digital frame and rejection rules

The 68-symbol header carries a 12-bit profile, CRC16 and six zero termination bits. Its convolutional encoder has constraint length 7 and octal generators $(171, 133)$, rate $1/2$; its 68 coded bits are repeated to 136 bits and mapped to 68 rotated-BPSK symbols. Terminated Viterbi decoding requires final state zero. CRC uses polynomial 0x1021, initial 0xFFFF, MSB-first integer fields. The body CRC covers the 13-bit length, source bits and known zero padding.

For $q = \log_2 M_{\mathrm{QAM}}$, $n = (N - 68)q$, $k = \lfloor rn \rfloor$ and source capacity $k - 29$. All 12 declared MCSs are constructor-admitted at both budgets. Sionna2.2.0 uses CPU FP32, one 5G-NR LDPC block and RV0: filler shortening, first-$2Z$ and tail puncturing, no repetition. Decoder settings are boxplus-phi/check, sum/variable, flooding, 20 iterations, LLR clipping 20, hard output and information bits returned, PCM pruning on, HARQ and early stopping off. Gray-mapped QAM has nominal $E_s = 2$ without per-frame renormalization. Public PCG64 scrambling follows rate matching; RX reverses its LLR signs. Its seed is derived from the first 16 SHA256 bytes, little-endian, of the canonical protocol/"scramble"/session/counter tuple. Exact mask and mapping definitions are in `ep_phy.py`; interleaver hashes and constructor metadata are in `phy_layouts.csv`.

N	q	r	k	n	BG	Z	Filler	Punctured
512	2	1/2	444	888	bg2	56	116	1908
512	2	2/3	592	888	bg2	72	128	2728
512	2	3/4	666	888	bg1	32	38	1250
512	2	5/6	740	888	bg1	36	52	1508
512	4	1/2	888	1776	bg2	96	72	3144
512	4	2/3	1184	1776	bg2	120	16	4448
512	4	3/4	1332	1776	bg1	64	76	2500
512	4	5/6	1480	1776	bg1	72	104	3016
512	6	1/2	1332	2664	bg2	144	108	4716
512	6	2/3	1776	2664	bg2	192	144	7176
512	6	3/4	1998	2664	bg1	96	114	3750
512	6	5/6	2220	2664	bg1	104	68	4340
1024	2	1/2	956	1912	bg2	96	4	3076
1024	2	2/3	1274	1912	bg2	128	6	4738
1024	2	3/4	1434	1912	bg1	72	150	2834
1024	2	5/6	1593	1912	bg1	80	167	3361
1024	4	1/2	1912	3824	bg2	192	8	6152
1024	4	2/3	2549	3824	bg2	256	11	9477
1024	4	3/4	2868	3824	bg1	144	300	5668
1024	4	5/6	3186	3824	bg1	160	334	6722
1024	6	1/2	2868	5736	bg2	288	12	9228
1024	6	2/3	3824	5736	bg2	384	16	14216
1024	6	3/4	4302	5736	bg1	208	274	8134
1024	6	5/6	4780	5736	bg1	224	148	9348

Profile CRC/catalogue, body CRC, invalid length, known padding or arithmetic canonical failures produce fixed gray RGB 0.5. A valid paid-NULL also produces gray, while preserving its full transmitted header/body and statistical denominator. For the current N1024 calibration, each method/MCS uses a public directory fixed before the pilot and shared across all target ceilings and source images. WHOLE has seven source profiles, NeST 651 and Ada 256, plus one paid-NULL profile in each directory. Thus every ID fits in 12 bits without changing the 68-symbol header. Under a

preconfigured policy, both terminals know the method, MCS and profile directory. The 12-bit profile identifies the actual endpoint within that directory; it does not identify a new directory or signal an MCS change. A policy change requires both terminals to configure the corresponding directory before sending data frames. The receiver reads the endpoint from the received header and uses the configured MCS for body demapping and decoding. Directory qualification performs two real complete-frame checks per directory (source and NULL), totaling 72 frames and 144 header/body decodes. Earlier studies used the joint endpoint/MCS catalogues identified below: within those shared catalogues, a profile entry also contained MCS parameters. Their catalogue identity remains part of the protocol context. The current U directories instead fix one MCS each.

Policy and directory index. The policy IDs below distinguish current calibration from earlier experiments. Entries are in WHOLE/NeST/Ada order, with $q = \log_2 M$; a single MCS entry applies to all listed methods. Ceilings (m, K) are target limits, not per-image transmitted endpoints; Ada's ceiling is a token count. Table IV gives current main policies, whereas the experiment index below identifies retained auxiliary configurations.

ID	N/design SNR	(q, r)	Ceilings W/N/A	Directory W/N/A
U4	1024/4	2,2/3	(7,0) / (7,25) / 256	U/U/U
U7	1024/7	4,1/2	(8,0) / (8,0) / 256	U/U/U
U10	1024/10	4,2/3	(9,0) / (8,126) / 256	U/U/U
U19	1024/19	$(6, 5/6); (6, 5/6); (4, 5/6)$	(10,0) / (10,0) / 256	U/U/U
H4	1024/4	2,2/3	(7,0) / (7,25) / 256	HW/HT/HA
H7	1024/7	4,1/2	(10,0) / (10,0) / 256	H7/HT/H7
H10	1024/10	4,2/3	(9,0) / (8,126) / 256	HW/HT/HA
H19	1024/19	6,5/6	(10,0) / (10,0) / 256	HW/HT/HA
L4	512/4	2,2/3	(6,0) / (5,27) / 256	LW/LT/LW
L10	512/10	4,2/3	(7,0) / (7,0) / 256	LW/LT/LW
L19	512/19	6,5/6	(8,0) / (8,84) / 256	LW/LT/LW
C1024	1024/10	4,2/3	(8,126)	HT
C512	512/10	4,2/3	(7,0)	LT

U denotes the public per-method/MCS directory used by unified calibration. Historical versions are identified by these SHA256 prefixes: HW `135d97cc`, HT `15d283c5`, HA `e867d196`, H7 `9130661f`, LW `d0b1ea52`, LT `06db78a9`, and KT `1e712673`. KT adds paid-NULL entries to the historical Token catalogue; existing source entries are unchanged. Full paths, hashes and the CSV index are in `inputs/policy_index_20261103/`.

Experiment	Policy IDs	Directory and scope
Current N1024 common500 and main Fig.3	U4/U7/U10/U19	U; Table IV
N512 common500	L4/L10/L19	LW/LT/LW; Table IV
Matched quality grids (S3)	C1024; C512	HT; LT; only allowed endpoints vary
Receiver A0/B0/C0 (S3)	H7/H10/H19, NeST ceiling	HT; Token prefixes; 19 dB is the final-scale structural check
Ada sampling/argmax (S3)	L4/L10/H10, Ada ceiling	LW/LW/HA; identical received prefixes
Equal information budgets (S4)	No MCS; ceilings (10,0)/(10,0)/256	Information-packet protocol; no PHY
Fixed 10-dB threshold test (S5)	H10 at actual 8–10 dB	HW/HT/HA; fixed transmitted frames
Supplementary 7-dB/10-dB examples (S6)	H7/H10	Historical outputs; main Fig.3 uses U10
Corrected-resource and compiled32 timing (S7)	H10	HW/HT/HA; compiled32 uses Ada entry
Matched-grid timing (S7)	C1024	Whole: HW; other grids: HT
Historical timing (S7)	H7/H10/L10	Earlier resource protocol, labeled in S7
Kodak24 (S8)	H4/H10/H19	WHOLE/Quarter: HW; Token: KT; Ada: HA

For Kodak, Quarter and Token share the NeST ceiling but use their respective endpoint grids. Receiver-only interventions reuse their original decoded prefixes. U10 and H10 have identical MCSs and target ceilings, so the H10 timing and threshold studies probe the same modulation/coding and source-limit choices; their distinct directory versions identify the actual frames used. At 19 dB, historical Ada H19 uses 64-QAM rate 5/6, while current U19 uses 16-QAM rate 5/6. Kodak therefore evaluates the historical frozen configurations. External-system manifests identify their unchanged configurations.

S1.4 Candidate pools and calibration

At N1024, all three token families use the same 12 MCSs, $q \in \{2, 4, 6\}$ and $r \in \{1/2, 2/3, 3/4, 5/6\}$. WHOLE has seven complete-scale target ceilings. NeST searches the 25 Quarter target ceilings and permits every integer endpoint up to each target. Ada has the maximum target of 256 and permits every shorter positive prefix. This gives 84/300/12 candidates for WHOLE/NeST/Ada, respectively. Equal MCS and selection rules do not mean identical endpoint grammars. NeST's search is finite over 25 ceilings, not an exhaustive search over all 651 possible ceilings.

Each candidate uses the same fixed 100 calibration sources with one channel realization (seed 4101) at 4/7/10/19 dB. Per family/SNR, the top three plus the predetermined maximum-ceiling QPSK rate-1/2 anchor advance, with duplicate candidate IDs removed. Final confirmation uses all 1000 calibration sources and three realizations (4101/4102/4103). Source-level means of DINOv2-L cosine include every gray failure and paid NULL; their equal-weight mean is maximized. Ties use a fixed lexical candidate-ID order. Identical actual streams or receiver inputs may be reused after full identity checks while all candidate logical positions remain in the denominator. The original serial Ada quality implementation, FP32/TF32-off configuration, CFG 4, T1, top-k 2000 and top-p 1 are fixed before evaluation.

The complete 1584 pilot candidate/SNR scores and all confirmation scores, with the public directory and policy hashes, are in `inputs/unified_calibration_20261103/`. No common500 result participates in policy selection. Frozen policies are then evaluated on 500 sources, three actual channel realizations per source (6201/6202/6203), yielding 18000 logical events. The original N512 calibration and its inherited Token refinement remain unchanged; the new matched N512 diagnostic below instead holds every factor except the termination grid fixed. Kodak, threshold and historical receiver diagnostics retain their specifically identified earlier policies.

The pilot contributes 158400 logical positions; confirmation contributes 141000. The confirmation candidate scores are:

Method	SNR	(q, r)	Ceiling	DINO cosine	Reject/NULL
Ada	4	2,1/2	(256, 0)	0.634400	0
Ada	4	2,2/3	(256, 0)	0.692317	0
Ada	4	2,3/4	(256, 0)	0.370409	1419
Ada	7	2,1/2	(256, 0)	0.634400	0
Ada	7	2,3/4	(256, 0)	0.714878	0
Ada	7	2,5/6	(256, 0)	0.733142	0
Ada	7	4,1/2	(256, 0)	0.769299	19
Ada	10	2,1/2	(256, 0)	0.634400	0
Ada	10	2,5/6	(256, 0)	0.733142	0
Ada	10	4,1/2	(256, 0)	0.774469	0
Ada	10	4,2/3	(256, 0)	0.821026	0
Ada	19	2,1/2	(256, 0)	0.634400	0
Ada	19	4,3/4	(256, 0)	0.831706	0
Ada	19	4,5/6	(256, 0)	0.833040	0
Ada	19	6,1/2	(256, 0)	0.831706	0
NeST	4	2,1/2	(10, 0)	0.519671	0
NeST	4	2,2/3	(7, 0)	0.586780	1
NeST	4	2,2/3	(7, 25)	0.586834	1
NeST	4	2,2/3	(7, 50)	0.586762	1
NeST	7	2,1/2	(10, 0)	0.519671	0
NeST	7	4,1/2	(8, 0)	0.681640	11
NeST	7	4,1/2	(8, 42)	0.681628	11
NeST	7	4,1/2	(8, 84)	0.681593	11
NeST	10	2,1/2	(10, 0)	0.519671	0
NeST	10	4,2/3	(8, 126)	0.754650	0
NeST	10	4,2/3	(9, 0)	0.754644	0

Method	SNR	(q, r)	Ceiling	DINO cosine	Reject/NULL
NeST	10	4,2/3	(9, 64)	0.754644	0
NeST	19	2,1/2	(10, 0)	0.519671	0
NeST	19	6,5/6	(9, 128)	0.829878	0
NeST	19	6,5/6	(9, 192)	0.830402	0
NeST	19	6,5/6	(10, 0)	0.830665	0
WHOLE	4	2,1/2	(10, 0)	0.456579	0
WHOLE	4	2,2/3	(7, 0)	0.518464	3
WHOLE	4	2,2/3	(8, 0)	0.518464	3
WHOLE	4	2,2/3	(9, 0)	0.518464	3
WHOLE	7	2,1/2	(10, 0)	0.456579	0
WHOLE	7	4,1/2	(8, 0)	0.648546	20
WHOLE	7	4,1/2	(9, 0)	0.648546	20
WHOLE	7	4,1/2	(10, 0)	0.648546	20
WHOLE	10	2,1/2	(10, 0)	0.456579	0
WHOLE	10	4,2/3	(8, 0)	0.690426	0
WHOLE	10	4,2/3	(9, 0)	0.690438	0
WHOLE	10	4,2/3	(10, 0)	0.690438	0
WHOLE	19	2,1/2	(10, 0)	0.456579	0
WHOLE	19	6,3/4	(9, 0)	0.804964	0
WHOLE	19	6,5/6	(9, 0)	0.819002	0
WHOLE	19	6,5/6	(10, 0)	0.819316	0

DINOv2 is the policy objective. DISTS, DreamSim and CLIP-I are feature-based source-reference measures; their agreement is not independent human perceptual validation. Source-capacity utilization excludes paid body fields and is not total air-interface efficiency. Ada may transmit all 256 tokens and leave capacity unused. N512 and N1024 are separately calibrated systems, not a budget-only causal intervention.

S1.5 Actual-length fitting and scan accounting

For each complete scale, the prior evaluates the probability map from known coarser maps. Raster tokens enter one shared incremental arithmetic encoder. At every allowed endpoint, a non-mutating finish produces a separate terminated stream. The current measured path is:

1. Fix MCS, public directory, legal endpoint grid and calibrated ceiling. Order endpoints by retained token count.
2. Traverse source tokens once, sharing probability evaluation and incremental coding. At each allowed endpoint copy its independently terminated bits without changing encoder state.
3. Compare every allowed candidate's actual length with the source capacity. For NeST/WHOLE require $b+29 \leq k$; Ada additionally pays its 10-bit class.
4. Select the most-token feasible endpoint below the ceiling. Transmit its entire terminated stream and paid profile; if none fits, transmit the paid NULL frame.
5. Recover the profile and then independently decode the corresponding body. Rejected/NULL images remain gray in every metric denominator.

For Quarter/Eighth, boundary counts are $\lfloor jL/d \rfloor$, $j = 1, \ldots, d-1$, $d = 4/8$, with duplicates removed and complete boundaries included. Token includes every integer boundary count. No partial stream is obtained by cutting a longer finalized stream.

Let E be emitted bits and P pending underflow bits. The original coder's finalization yields $b = E+P+2$. Every E1/E2 renormalization emits $P+1$ bits and clears P, and every E3 increments P; each raises $E+P$ by one. A token's interval update alone leaves it unchanged. Thus independently terminated lengths along this shared path are nondecreasing, with possible ties. We checked the exact original integer coder on 512 synthetic prefixes, and 976500 existing streams from cal1000/common500 with all 975000 adjacent length relations. Array hashes, non-mutating finish and canonical roundtrips pass; no image or PHY was rerun for this audit.

The timed code still generates and scans all allowed candidates. Its historical `length_checks` field is the first feasible rank, not the list comprehension's full comparison count. Candidate comparisons are not extra prior forwards. Monotonic lookup or early stopping could reduce work but is not implemented or retimed here; no such speedup is attributed to the reported timings.

S1.6 Reference vectors and verification

Six deterministic CPU probability cases and six independently terminated prefixes verify positive integer frequencies, total frequency, first-index ties, non-mutating finalization, independent decoding, exact 30-bit extension and rejection of noncanonical appended bits. Run from the repository root:

```
python experiments/nestcom-v9-completion-20261101/verify_cdf_ac.py
-repo .  -output /tmp/nest-cdf-check
```

The output directory must be fresh. Two real-model vectors below record complete TX/RX scale-CDF hashes, exact bits/tokens, canonical checks, model/numerical identity and source provenance. RX did not consume TX probability tables, source tokens or hidden source states. Comparisons were made only after independent reception completed.

Vector	(m, K)	Tokens	Source bits	CDF scales	Exact TX/RX
Whole	(4, 0)	30	318	4	yes
Boundary	(6, 34)	125	1241	7	yes

The whole reference uses one explicitly counted receiver diagnostic without a new PHY event; the boundary reference reuses a completed Kodak input. Integer vectors replay without weights. Independent model/CDF regeneration requires the pinned assets and numerical environment; reference hashes characterize that tested route.

Common500 uses its original 10 000×500 source-bootstrap bank; controls use a separately ordered 10 000×100 bank. Kodak uses the existing 10 000×24 bank. Each source first averages its three events, then receives equal weight; paired differences precede resampling. Full source IDs, pixel hashes and native receiver identities control reuse. No simultaneous multiplicity-adjusted interval claim is made.

S2 Complete quality results

The N1024 WHOLE/NeST/Ada rows below use the new uniformly calibrated policies. Latent JSCC, SwinJSCC, BPG and the supplementary ARPC digital adaptation reuse their exact original native outputs after source-ID, preprocessing, pixel-reference and metric checks. The three newly calibrated token methods share the source/repetition noise vector. Reused external methods retain their original independent noise identities; cross-method pairing is by source, not a claim of shared noise across all systems. N512 rows retain the independently calibrated study. Every source averages three events before equal-source means or direct paired differences. Intervals use the same fixed 10000-by-500 source bank, with no simultaneous-coverage claim.

N	SNR	Method	Metric	Mean [pointwise 95% CI]
1024	4	ARPC	PSNR	+14.18407 [+13.95318, +14.41624]
1024	4	ARPC	LPIPS	+0.51240 [+0.50465, +0.52028]
1024	4	ARPC	DreamSim	+0.50098 [+0.49060, +0.51135]
1024	4	ARPC	CLIP-I	+0.64417 [+0.63589, +0.65219]
1024	4	ARPC	DISTS	+0.34360 [+0.33933, +0.34804]
1024	7	ARPC	PSNR	+14.75886 [+14.53731, +14.98569]
1024	7	ARPC	LPIPS	+0.48086 [+0.47449, +0.48726]
1024	7	ARPC	DreamSim	+0.47514 [+0.46502, +0.48512]
1024	7	ARPC	CLIP-I	+0.65078 [+0.64272, +0.65870]
1024	7	ARPC	DISTS	+0.32867 [+0.32493, +0.33247]
1024	10	ARPC	PSNR	+16.47078 [+16.23791, +16.71812]
1024	10	ARPC	LPIPS	+0.34127 [+0.33589, +0.34671]

N	SNR	Method	Metric	Mean [pointwise 95% CI]
1024	10	ARPC	DreamSim	+0.30617 [+0.29795, +0.31468]
1024	10	ARPC	CLIP-I	+0.71336 [+0.70484, +0.72177]
1024	10	ARPC	DISTS	+0.22777 [+0.22565, +0.22991]
1024	19	ARPC	PSNR	+18.19870 [+17.94205, +18.46612]
1024	19	ARPC	LPIPS	+0.25034 [+0.24504, +0.25564]
1024	19	ARPC	DreamSim	+0.19520 [+0.18868, +0.20196]
1024	19	ARPC	CLIP-I	+0.78799 [+0.77986, +0.79620]
1024	19	ARPC	DISTS	+0.18119 [+0.17910, +0.18329]
1024	4	BPG	PSNR	+20.74259 [+20.48394, +21.01048]
1024	4	BPG	LPIPS	+0.72548 [+0.71176, +0.73898]
1024	4	BPG	DreamSim	+0.57367 [+0.56326, +0.58404]
1024	4	BPG	CLIP-I	+0.58586 [+0.57844, +0.59343]
1024	4	BPG	DISTS	+0.40221 [+0.39859, +0.40577]
1024	7	BPG	PSNR	+21.71706 [+21.43692, +22.00492]
1024	7	BPG	LPIPS	+0.64997 [+0.63567, +0.66440]
1024	7	BPG	DreamSim	+0.47885 [+0.46815, +0.48964]
1024	7	BPG	CLIP-I	+0.61501 [+0.60661, +0.62317]
1024	7	BPG	DISTS	+0.36759 [+0.36373, +0.37158]
1024	10	BPG	PSNR	+22.54299 [+22.25423, +22.84398]
1024	10	BPG	LPIPS	+0.58937 [+0.57491, +0.60402]
1024	10	BPG	DreamSim	+0.41655 [+0.40602, +0.42735]
1024	10	BPG	CLIP-I	+0.63501 [+0.62685, +0.64313]
1024	10	BPG	DISTS	+0.34088 [+0.33739, +0.34443]
1024	19	BPG	PSNR	+24.23012 [+23.91237, +24.55901]
1024	19	BPG	LPIPS	+0.45839 [+0.44408, +0.47289]
1024	19	BPG	DreamSim	+0.29638 [+0.28691, +0.30623]
1024	19	BPG	CLIP-I	+0.68431 [+0.67584, +0.69274]
1024	19	BPG	DISTS	+0.29196 [+0.28819, +0.29569]
1024	4	Latent	PSNR	+19.65784 [+19.44190, +19.88023]
1024	4	Latent	LPIPS	+0.21878 [+0.21455, +0.22299]
1024	4	Latent	DreamSim	+0.23168 [+0.22524, +0.23825]
1024	4	Latent	CLIP-I	+0.74083 [+0.73298, +0.74851]
1024	4	Latent	DISTS	+0.22180 [+0.21932, +0.22431]
1024	7	Latent	PSNR	+20.36893 [+20.14132, +20.60239]
1024	7	Latent	LPIPS	+0.18928 [+0.18537, +0.19319]
1024	7	Latent	DreamSim	+0.19117 [+0.18547, +0.19693]
1024	7	Latent	CLIP-I	+0.77169 [+0.76408, +0.77913]
1024	7	Latent	DISTS	+0.20695 [+0.20442, +0.20950]
1024	10	Latent	PSNR	+20.80259 [+20.56741, +21.04386]
1024	10	Latent	LPIPS	+0.17326 [+0.16950, +0.17697]
1024	10	Latent	DreamSim	+0.17115 [+0.16587, +0.17653]
1024	10	Latent	CLIP-I	+0.78918 [+0.78168, +0.79654]
1024	10	Latent	DISTS	+0.19864 [+0.19606, +0.20124]
1024	19	Latent	PSNR	+21.24461 [+21.00138, +21.49488]
1024	19	Latent	LPIPS	+0.15848 [+0.15491, +0.16204]
1024	19	Latent	DreamSim	+0.15460 [+0.14960, +0.15965]
1024	19	Latent	CLIP-I	+0.80414 [+0.79651, +0.81159]
1024	19	Latent	DISTS	+0.19110 [+0.18847, +0.19376]
1024	4	Swin	PSNR	+21.67102 [+21.41910, +21.93451]
1024	4	Swin	LPIPS	+0.52973 [+0.51727, +0.54219]
1024	4	Swin	DreamSim	+0.43591 [+0.42657, +0.44546]
1024	4	Swin	CLIP-I	+0.62668 [+0.61916, +0.63428]
1024	4	Swin	DISTS	+0.34884 [+0.34588, +0.35185]
1024	7	Swin	PSNR	+22.47464 [+22.21420, +22.74510]
1024	7	Swin	LPIPS	+0.49334 [+0.48081, +0.50593]
1024	7	Swin	DreamSim	+0.38872 [+0.37959, +0.39807]
1024	7	Swin	CLIP-I	+0.65049 [+0.64287, +0.65827]

N	SNR	Method	Metric	Mean [pointwise 95% CI]
1024	7	Swin	DISTS	+0.33363 [+0.33067, +0.33664]
1024	10	Swin	PSNR	+23.02595 [+22.75708, +23.30447]
1024	10	Swin	LPIPS	+0.46866 [+0.45613, +0.48130]
1024	10	Swin	DreamSim	+0.35639 [+0.34736, +0.36558]
1024	10	Swin	CLIP-I	+0.66908 [+0.66126, +0.67681]
1024	10	Swin	DISTS	+0.32239 [+0.31935, +0.32544]
1024	19	Swin	PSNR	+23.69820 [+23.41493, +23.98647]
1024	19	Swin	LPIPS	+0.44003 [+0.42754, +0.45280]
1024	19	Swin	DreamSim	+0.31795 [+0.30924, +0.32685]
1024	19	Swin	CLIP-I	+0.69660 [+0.68874, +0.70444]
1024	19	Swin	DISTS	+0.30806 [+0.30490, +0.31123]
1024	4	WHOLE	PSNR	+16.83227 [+16.60005, +17.07610]
1024	4	WHOLE	LPIPS	+0.28783 [+0.28187, +0.29378]
1024	4	WHOLE	DreamSim	+0.20460 [+0.19737, +0.21184]
1024	4	WHOLE	CLIP-I	+0.80770 [+0.79894, +0.81618]
1024	4	WHOLE	DISTS	+0.20283 [+0.20033, +0.20541]
1024	7	WHOLE	PSNR	+18.50839 [+18.26293, +18.76356]
1024	7	WHOLE	LPIPS	+0.22434 [+0.21821, +0.23042]
1024	7	WHOLE	DreamSim	+0.15276 [+0.14644, +0.15931]
1024	7	WHOLE	CLIP-I	+0.84501 [+0.83683, +0.85266]
1024	7	WHOLE	DISTS	+0.17740 [+0.17390, +0.18112]
1024	10	WHOLE	PSNR	+19.31282 [+19.02087, +19.60883]
1024	10	WHOLE	LPIPS	+0.19872 [+0.19218, +0.20525]
1024	10	WHOLE	DreamSim	+0.13172 [+0.12594, +0.13763]
1024	10	WHOLE	CLIP-I	+0.85774 [+0.84973, +0.86557]
1024	10	WHOLE	DISTS	+0.16213 [+0.15914, +0.16511]
1024	19	WHOLE	PSNR	+21.84437 [+21.57627, +22.12225]
1024	19	WHOLE	LPIPS	+0.12338 [+0.11983, +0.12691]
1024	19	WHOLE	DreamSim	+0.08150 [+0.07852, +0.08467]
1024	19	WHOLE	CLIP-I	+0.89687 [+0.89117, +0.90238]
1024	19	WHOLE	DISTS	+0.12744 [+0.12526, +0.12972]
1024	4	NeST	PSNR	+17.64940 [+17.40096, +17.91029]
1024	4	NeST	LPIPS	+0.25544 [+0.24922, +0.26159]
1024	4	NeST	DreamSim	+0.17486 [+0.16822, +0.18169]
1024	4	NeST	CLIP-I	+0.82686 [+0.81808, +0.83519]
1024	4	NeST	DISTS	+0.18865 [+0.18603, +0.19135]
1024	7	NeST	PSNR	+19.00602 [+18.75004, +19.27588]
1024	7	NeST	LPIPS	+0.20578 [+0.19989, +0.21174]
1024	7	NeST	DreamSim	+0.13688 [+0.13119, +0.14276]
1024	7	NeST	CLIP-I	+0.85495 [+0.84729, +0.86247]
1024	7	NeST	DISTS	+0.16709 [+0.16386, +0.17049]
1024	10	NeST	PSNR	+20.19404 [+19.92706, +20.46997]
1024	10	NeST	LPIPS	+0.16579 [+0.16126, +0.17040]
1024	10	NeST	DreamSim	+0.10476 [+0.10099, +0.10873]
1024	10	NeST	CLIP-I	+0.87692 [+0.87020, +0.88351]
1024	10	NeST	DISTS	+0.14806 [+0.14572, +0.15049]
1024	19	NeST	PSNR	+22.33412 [+22.05308, +22.62458]
1024	19	NeST	LPIPS	+0.11513 [+0.11146, +0.11875]
1024	19	NeST	DreamSim	+0.07719 [+0.07425, +0.08028]
1024	19	NeST	CLIP-I	+0.90149 [+0.89605, +0.90685]
1024	19	NeST	DISTS	+0.12262 [+0.12033, +0.12501]
1024	4	Ada	PSNR	+13.33209 [+13.12041, +13.55321]
1024	4	Ada	LPIPS	+0.39520 [+0.38782, +0.40263]
1024	4	Ada	DreamSim	+0.24310 [+0.23346, +0.25323]
1024	4	Ada	CLIP-I	+0.82801 [+0.81999, +0.83582]
1024	4	Ada	DISTS	+0.23003 [+0.22634, +0.23382]
1024	7	Ada	PSNR	+15.92903 [+15.65016, +16.21807]

N	SNR	Method	Metric	Mean [pointwise 95% CI]
1024	7	Ada	LPIPS	+0.27056 [+0.26296, +0.27849]
1024	7	Ada	DreamSim	+0.14207 [+0.13514, +0.14930]
1024	7	Ada	CLIP-I	+0.85907 [+0.85189, +0.86617]
1024	7	Ada	DISTS	+0.17783 [+0.17363, +0.18220]
1024	10	Ada	PSNR	+19.55896 [+19.25831, +19.85439]
1024	10	Ada	LPIPS	+0.16735 [+0.16180, +0.17305]
1024	10	Ada	DreamSim	+0.08710 [+0.08352, +0.09089]
1024	10	Ada	CLIP-I	+0.88413 [+0.87777, +0.89041]
1024	10	Ada	DISTS	+0.12751 [+0.12507, +0.13006]
1024	19	Ada	PSNR	+20.93405 [+20.67159, +21.20482]
1024	19	Ada	LPIPS	+0.13769 [+0.13387, +0.14149]
1024	19	Ada	DreamSim	+0.07626 [+0.07337, +0.07926]
1024	19	Ada	CLIP-I	+0.89240 [+0.88639, +0.89829]
1024	19	Ada	DISTS	+0.11363 [+0.11195, +0.11536]
512	4	Ada	PSNR	+11.31831 [+11.12297, +11.51854]
512	4	Ada	LPIPS	+0.55416 [+0.54659, +0.56167]
512	4	Ada	DreamSim	+0.38473 [+0.37293, +0.39657]
512	4	Ada	CLIP-I	+0.78921 [+0.78074, +0.79763]
512	4	Ada	DISTS	+0.29961 [+0.29541, +0.30391]
512	10	Ada	PSNR	+13.16468 [+12.94332, +13.39518]
512	10	Ada	LPIPS	+0.41098 [+0.40364, +0.41835]
512	10	Ada	DreamSim	+0.25817 [+0.24805, +0.26846]
512	10	Ada	CLIP-I	+0.82575 [+0.81785, +0.83350]
512	10	Ada	DISTS	+0.23671 [+0.23306, +0.24043]
512	19	Ada	PSNR	+17.69003 [+17.38737, +17.99744]
512	19	Ada	LPIPS	+0.21457 [+0.20778, +0.22136]
512	19	Ada	DreamSim	+0.10775 [+0.10289, +0.11287]
512	19	Ada	CLIP-I	+0.87228 [+0.86544, +0.87899]
512	19	Ada	DISTS	+0.14958 [+0.14661, +0.15261]
512	4	WHOLE	PSNR	+15.10655 [+14.85320, +15.36607]
512	4	WHOLE	LPIPS	+0.39512 [+0.38531, +0.40493]
512	4	WHOLE	DreamSim	+0.32466 [+0.31308, +0.33661]
512	4	WHOLE	CLIP-I	+0.74694 [+0.73701, +0.75681]
512	4	WHOLE	DISTS	+0.25800 [+0.25370, +0.26247]
512	10	WHOLE	PSNR	+16.82131 [+16.59007, +17.06388]
512	10	WHOLE	LPIPS	+0.28808 [+0.28216, +0.29399]
512	10	WHOLE	DreamSim	+0.20486 [+0.19764, +0.21207]
512	10	WHOLE	CLIP-I	+0.80744 [+0.79868, +0.81586]
512	10	WHOLE	DISTS	+0.20297 [+0.20052, +0.20552]
512	19	WHOLE	PSNR	+18.63777 [+18.38451, +18.90215]
512	19	WHOLE	LPIPS	+0.21754 [+0.21220, +0.22296]
512	19	WHOLE	DreamSim	+0.14580 [+0.14038, +0.15132]
512	19	WHOLE	CLIP-I	+0.84835 [+0.84040, +0.85602]
512	19	WHOLE	DISTS	+0.17184 [+0.16935, +0.17438]
512	4	NeST	PSNR	+15.61081 [+15.37367, +15.85747]
512	4	NeST	LPIPS	+0.35654 [+0.34905, +0.36393]
512	4	NeST	DreamSim	+0.27405 [+0.26514, +0.28318]
512	4	NeST	CLIP-I	+0.76914 [+0.75969, +0.77862]
512	4	NeST	DISTS	+0.23629 [+0.23302, +0.23961]
512	10	NeST	PSNR	+17.41292 [+17.16882, +17.66911]
512	10	NeST	LPIPS	+0.26501 [+0.25877, +0.27119]
512	10	NeST	DreamSim	+0.18374 [+0.17689, +0.19076]
512	10	NeST	CLIP-I	+0.82109 [+0.81231, +0.82950]
512	10	NeST	DISTS	+0.19216 [+0.18960, +0.19472]
512	19	NeST	PSNR	+19.63169 [+19.36599, +19.90637]
512	19	NeST	LPIPS	+0.18324 [+0.17819, +0.18829]
512	19	NeST	DreamSim	+0.11499 [+0.11062, +0.11957]

N	SNR	Method	Metric	Mean [pointwise 95% CI]
512	19	NeST	CLIP-I	+0.86889 [+0.86183, +0.87572]
512	19	NeST	DISTS	+0.15539 [+0.15295, +0.15787]
512	4	Quarter	PSNR	+15.49872 [+15.26339, +15.74203]
512	4	Quarter	LPIPS	+0.36517 [+0.35750, +0.37269]
512	4	Quarter	DreamSim	+0.28187 [+0.27274, +0.29107]
512	4	Quarter	CLIP-I	+0.76569 [+0.75648, +0.77489]
512	4	Quarter	DISTS	+0.24062 [+0.23738, +0.24393]
512	10	Quarter	PSNR	+17.21988 [+16.97768, +17.47598]
512	10	Quarter	LPIPS	+0.27260 [+0.26625, +0.27880]
512	10	Quarter	DreamSim	+0.19217 [+0.18495, +0.19957]
512	10	Quarter	CLIP-I	+0.81498 [+0.80623, +0.82341]
512	10	Quarter	DISTS	+0.19583 [+0.19312, +0.19862]
512	19	Quarter	PSNR	+19.42978 [+19.16706, +19.70150]
512	19	Quarter	LPIPS	+0.19005 [+0.18497, +0.19519]
512	19	Quarter	DreamSim	+0.12168 [+0.11686, +0.12663]
512	19	Quarter	CLIP-I	+0.86400 [+0.85678, +0.87100]
512	19	Quarter	DISTS	+0.15865 [+0.15625, +0.16113]

N	SNR	Comparison	Metric	Difference [paired 95% CI]
1024	4	NeST-WHOLE	PSNR	+0.81713 [+0.75992, +0.87680]
1024	4	NeST-WHOLE	LPIPS	-0.03239 [-0.03425, -0.03040]
1024	4	NeST-WHOLE	DreamSim	-0.02974 [-0.03303, -0.02644]
1024	4	NeST-WHOLE	CLIP-I	+0.01916 [+0.01528, +0.02297]
1024	4	NeST-WHOLE	DISTS	-0.01418 [-0.01552, -0.01265]
1024	7	NeST-WHOLE	PSNR	+0.49764 [+0.44148, +0.55287]
1024	7	NeST-WHOLE	LPIPS	-0.01856 [-0.02281, -0.01443]
1024	7	NeST-WHOLE	DreamSim	-0.01588 [-0.02072, -0.01117]
1024	7	NeST-WHOLE	CLIP-I	+0.00994 [+0.00652, +0.01350]
1024	7	NeST-WHOLE	DISTS	-0.01031 [-0.01391, -0.00683]
1024	10	NeST-WHOLE	PSNR	+0.88122 [+0.80761, +0.95550]
1024	10	NeST-WHOLE	LPIPS	-0.03293 [-0.03561, -0.03021]
1024	10	NeST-WHOLE	DreamSim	-0.02697 [-0.02990, -0.02408]
1024	10	NeST-WHOLE	CLIP-I	+0.01918 [+0.01551, +0.02307]
1024	10	NeST-WHOLE	DISTS	-0.01407 [-0.01532, -0.01281]
1024	19	NeST-WHOLE	PSNR	+0.48975 [+0.45749, +0.52298]
1024	19	NeST-WHOLE	LPIPS	-0.00824 [-0.00868, -0.00782]
1024	19	NeST-WHOLE	DreamSim	-0.00431 [-0.00508, -0.00359]
1024	19	NeST-WHOLE	CLIP-I	+0.00462 [+0.00302, +0.00624]
1024	19	NeST-WHOLE	DISTS	-0.00482 [-0.00532, -0.00432]
1024	4	NeST-Ada	PSNR	+4.31731 [+4.12308, +4.51801]
1024	4	NeST-Ada	LPIPS	-0.13975 [-0.14538, -0.13426]
1024	4	NeST-Ada	DreamSim	-0.06824 [-0.07605, -0.06039]
1024	4	NeST-Ada	CLIP-I	-0.00115 [-0.00740, +0.00494]
1024	4	NeST-Ada	DISTS	-0.04138 [-0.04464, -0.03809]
1024	7	NeST-Ada	PSNR	+3.07700 [+2.88114, +3.28063]
1024	7	NeST-Ada	LPIPS	-0.06479 [-0.07023, -0.05944]
1024	7	NeST-Ada	DreamSim	-0.00519 [-0.01159, +0.00101]
1024	7	NeST-Ada	CLIP-I	-0.00412 [-0.00946, +0.00107]
1024	7	NeST-Ada	DISTS	-0.01074 [-0.01491, -0.00670]
1024	10	NeST-Ada	PSNR	+0.63508 [+0.52112, +0.75347]
1024	10	NeST-Ada	LPIPS	-0.00156 [-0.00407, +0.00081]
1024	10	NeST-Ada	DreamSim	+0.01766 [+0.01532, +0.01994]
1024	10	NeST-Ada	CLIP-I	-0.00721 [-0.01140, -0.00307]
1024	10	NeST-Ada	DISTS	+0.02055 [+0.01885, +0.02218]
1024	19	NeST-Ada	PSNR	+1.40008 [+1.34001, +1.45887]

N	SNR	Comparison	Metric	Difference [paired 95% CI]
1024	19	NeST-Ada	LPIPS	-0.02255 [-0.02356, -0.02153]
1024	19	NeST-Ada	DreamSim	+0.00093 [-0.00087, +0.00276]
1024	19	NeST-Ada	CLIP-I	+0.00909 [+0.00528, +0.01298]
1024	19	NeST-Ada	DISTS	+0.00899 [+0.00756, +0.01045]
1024	4	WHOLE-Ada	PSNR	+3.50018 [+3.31221, +3.69540]
1024	4	WHOLE-Ada	LPIPS	-0.10736 [-0.11317, -0.10169]
1024	4	WHOLE-Ada	DreamSim	-0.03850 [-0.04715, -0.03006]
1024	4	WHOLE-Ada	CLIP-I	-0.02031 [-0.02711, -0.01385]
1024	4	WHOLE-Ada	DISTS	-0.02720 [-0.03044, -0.02398]
1024	7	WHOLE-Ada	PSNR	+2.57936 [+2.37823, +2.78822]
1024	7	WHOLE-Ada	LPIPS	-0.04623 [-0.05176, -0.04078]
1024	7	WHOLE-Ada	DreamSim	+0.01069 [+0.00437, +0.01710]
1024	7	WHOLE-Ada	CLIP-I	-0.01406 [-0.01971, -0.00854]
1024	7	WHOLE-Ada	DISTS	-0.00043 [-0.00464, +0.00372]
1024	10	WHOLE-Ada	PSNR	-0.24614 [-0.37122, -0.11750]
1024	10	WHOLE-Ada	LPIPS	+0.03137 [+0.02789, +0.03480]
1024	10	WHOLE-Ada	DreamSim	+0.04462 [+0.04085, +0.04841]
1024	10	WHOLE-Ada	CLIP-I	-0.02640 [-0.03148, -0.02151]
1024	10	WHOLE-Ada	DISTS	+0.03462 [+0.03271, +0.03645]
1024	19	WHOLE-Ada	PSNR	+0.91033 [+0.85993, +0.96070]
1024	19	WHOLE-Ada	LPIPS	-0.01431 [-0.01529, -0.01333]
1024	19	WHOLE-Ada	DreamSim	+0.00524 [+0.00344, +0.00705]
1024	19	WHOLE-Ada	CLIP-I	+0.00447 [+0.00080, +0.00831]
1024	19	WHOLE-Ada	DISTS	+0.01381 [+0.01249, +0.01516]
1024	4	NeST-Latent	PSNR	-2.00844 [-2.10404, -1.91048]
1024	4	NeST-Latent	LPIPS	+0.03666 [+0.03352, +0.03988]
1024	4	NeST-Latent	DreamSim	-0.05681 [-0.06208, -0.05142]
1024	4	NeST-Latent	CLIP-I	+0.08603 [+0.07938, +0.09230]
1024	4	NeST-Latent	DISTS	-0.03315 [-0.03528, -0.03098]
1024	7	NeST-Latent	PSNR	-1.36290 [-1.45842, -1.26437]
1024	7	NeST-Latent	LPIPS	+0.01650 [+0.01306, +0.02020]
1024	7	NeST-Latent	DreamSim	-0.05429 [-0.05900, -0.04946]
1024	7	NeST-Latent	CLIP-I	+0.08326 [+0.07756, +0.08888]
1024	7	NeST-Latent	DISTS	-0.03986 [-0.04262, -0.03696]
1024	10	NeST-Latent	PSNR	-0.60855 [-0.69268, -0.51994]
1024	10	NeST-Latent	LPIPS	-0.00746 [-0.00940, -0.00555]
1024	10	NeST-Latent	DreamSim	-0.06639 [-0.06971, -0.06317]
1024	10	NeST-Latent	CLIP-I	+0.08774 [+0.08238, +0.09314]
1024	10	NeST-Latent	DISTS	-0.05058 [-0.05219, -0.04899]
1024	19	NeST-Latent	PSNR	+1.08951 [+0.99436, +1.18423]
1024	19	NeST-Latent	LPIPS	-0.04335 [-0.04493, -0.04170]
1024	19	NeST-Latent	DreamSim	-0.07741 [-0.08074, -0.07422]
1024	19	NeST-Latent	CLIP-I	+0.09735 [+0.09219, +0.10272]
1024	19	NeST-Latent	DISTS	-0.06847 [-0.07005, -0.06689]
1024	4	NeST-Swin	PSNR	-4.02162 [-4.13349, -3.90899]
1024	4	NeST-Swin	LPIPS	-0.27428 [-0.28476, -0.26443]
1024	4	NeST-Swin	DreamSim	-0.26105 [-0.26981, -0.25244]
1024	4	NeST-Swin	CLIP-I	+0.20018 [+0.19222, +0.20796]
1024	4	NeST-Swin	DISTS	-0.16019 [-0.16323, -0.15707]
1024	7	NeST-Swin	PSNR	-3.46862 [-3.57271, -3.36560]
1024	7	NeST-Swin	LPIPS	-0.28757 [-0.29819, -0.27739]
1024	7	NeST-Swin	DreamSim	-0.25184 [-0.25993, -0.24370]
1024	7	NeST-Swin	CLIP-I	+0.20446 [+0.19714, +0.21171]
1024	7	NeST-Swin	DISTS	-0.16655 [-0.17001, -0.16289]
1024	10	NeST-Swin	PSNR	-2.83191 [-2.91889, -2.74662]
1024	10	NeST-Swin	LPIPS	-0.30286 [-0.31304, -0.29318]
1024	10	NeST-Swin	DreamSim	-0.25163 [-0.25921, -0.24423]

N	SNR	Comparison	Metric	Difference [paired 95% CI]
1024	10	NeST-Swin	CLIP-I	+0.20784 [+0.20088, +0.21498]
1024	10	NeST-Swin	DISTS	-0.17433 [-0.17689, -0.17179]
1024	19	NeST-Swin	PSNR	-1.36408 [-1.44263, -1.28515]
1024	19	NeST-Swin	LPIPS	-0.32490 [-0.33497, -0.31516]
1024	19	NeST-Swin	DreamSim	-0.24076 [-0.24816, -0.23368]
1024	19	NeST-Swin	CLIP-I	+0.20489 [+0.19822, +0.21150]
1024	19	NeST-Swin	DISTS	-0.18544 [-0.18795, -0.18291]
1024	4	NeST-BPG	PSNR	-3.09319 [-3.19523, -2.99035]
1024	4	NeST-BPG	LPIPS	-0.47003 [-0.48044, -0.45979]
1024	4	NeST-BPG	DreamSim	-0.39881 [-0.40911, -0.38847]
1024	4	NeST-BPG	CLIP-I	+0.24100 [+0.23208, +0.24991]
1024	4	NeST-BPG	DISTS	-0.21356 [-0.21684, -0.21023]
1024	7	NeST-BPG	PSNR	-2.71104 [-2.82425, -2.59383]
1024	7	NeST-BPG	LPIPS	-0.44419 [-0.45581, -0.43286]
1024	7	NeST-BPG	DreamSim	-0.34197 [-0.35196, -0.33225]
1024	7	NeST-BPG	CLIP-I	+0.23994 [+0.23141, +0.24872]
1024	7	NeST-BPG	DISTS	-0.20050 [-0.20461, -0.19633]
1024	10	NeST-BPG	PSNR	-2.34895 [-2.43360, -2.26426]
1024	10	NeST-BPG	LPIPS	-0.42358 [-0.43505, -0.41248]
1024	10	NeST-BPG	DreamSim	-0.31179 [-0.32132, -0.30255]
1024	10	NeST-BPG	CLIP-I	+0.24191 [+0.23400, +0.24964]
1024	10	NeST-BPG	DISTS	-0.19282 [-0.19556, -0.19008]
1024	19	NeST-BPG	PSNR	-1.89600 [-1.97754, -1.81607]
1024	19	NeST-BPG	LPIPS	-0.34325 [-0.35483, -0.33196]
1024	19	NeST-BPG	DreamSim	-0.21919 [-0.22742, -0.21119]
1024	19	NeST-BPG	CLIP-I	+0.21718 [+0.20920, +0.22538]
1024	19	NeST-BPG	DISTS	-0.16934 [-0.17197, -0.16666]
1024	4	NeST-ARPC	PSNR	+3.46533 [+3.32232, +3.61180]
1024	4	NeST-ARPC	LPIPS	-0.25696 [-0.26356, -0.25057]
1024	4	NeST-ARPC	DreamSim	-0.32611 [-0.33635, -0.31576]
1024	4	NeST-ARPC	CLIP-I	+0.18269 [+0.17469, +0.19080]
1024	4	NeST-ARPC	DISTS	-0.15495 [-0.15951, -0.15047]
1024	7	NeST-ARPC	PSNR	+4.24716 [+4.10799, +4.39003]
1024	7	NeST-ARPC	LPIPS	-0.27508 [-0.28062, -0.26963]
1024	7	NeST-ARPC	DreamSim	-0.33827 [-0.34804, -0.32841]
1024	7	NeST-ARPC	CLIP-I	+0.20416 [+0.19639, +0.21196]
1024	7	NeST-ARPC	DISTS	-0.16158 [-0.16590, -0.15720]
1024	10	NeST-ARPC	PSNR	+3.72326 [+3.61522, +3.83454]
1024	10	NeST-ARPC	LPIPS	-0.17547 [-0.17876, -0.17216]
1024	10	NeST-ARPC	DreamSim	-0.20141 [-0.20900, -0.19388]
1024	10	NeST-ARPC	CLIP-I	+0.16356 [+0.15623, +0.17083]
1024	10	NeST-ARPC	DISTS	-0.07971 [-0.08218, -0.07719]
1024	19	NeST-ARPC	PSNR	+4.13543 [+4.03989, +4.22803]
1024	19	NeST-ARPC	LPIPS	-0.13520 [-0.13846, -0.13205]
1024	19	NeST-ARPC	DreamSim	-0.11801 [-0.12370, -0.11237]
1024	19	NeST-ARPC	CLIP-I	+0.11350 [+0.10702, +0.11995]
1024	19	NeST-ARPC	DISTS	-0.05857 [-0.06078, -0.05638]
512	4	NeST-Quarter	PSNR	+0.11210 [+0.06798, +0.16080]
512	4	NeST-Quarter	LPIPS	-0.00863 [-0.01104, -0.00621]
512	4	NeST-Quarter	DreamSim	-0.00782 [-0.01132, -0.00447]
512	4	NeST-Quarter	CLIP-I	+0.00345 [-0.00043, +0.00738]
512	4	NeST-Quarter	DISTS	-0.00433 [-0.00627, -0.00247]
512	10	NeST-Quarter	PSNR	+0.19304 [+0.16224, +0.22489]
512	10	NeST-Quarter	LPIPS	-0.00759 [-0.00914, -0.00622]
512	10	NeST-Quarter	DreamSim	-0.00842 [-0.01061, -0.00630]
512	10	NeST-Quarter	CLIP-I	+0.00611 [+0.00330, +0.00898]
512	10	NeST-Quarter	DISTS	-0.00367 [-0.00476, -0.00270]

N	SNR	Comparison	Metric	Difference [paired 95% CI]
512	19	NeST-Quarter	PSNR	+0.20191 [+0.17590, +0.22869]
512	19	NeST-Quarter	LPIPS	-0.00681 [-0.00771, -0.00595]
512	19	NeST-Quarter	DreamSim	-0.00669 [-0.00828, -0.00521]
512	19	NeST-Quarter	CLIP-I	+0.00489 [+0.00239, +0.00738]
512	19	NeST-Quarter	DISTS	-0.00326 [-0.00391, -0.00263]
512	4	NeST-WHOLE	PSNR	+0.50427 [+0.43976, +0.57335]
512	4	NeST-WHOLE	LPIPS	-0.03858 [-0.04275, -0.03445]
512	4	NeST-WHOLE	DreamSim	-0.05062 [-0.05665, -0.04454]
512	4	NeST-WHOLE	CLIP-I	+0.02219 [+0.01714, +0.02740]
512	4	NeST-WHOLE	DISTS	-0.02171 [-0.02449, -0.01904]
512	10	NeST-WHOLE	PSNR	+0.59161 [+0.54369, +0.64265]
512	10	NeST-WHOLE	LPIPS	-0.02307 [-0.02452, -0.02161]
512	10	NeST-WHOLE	DreamSim	-0.02112 [-0.02363, -0.01865]
512	10	NeST-WHOLE	CLIP-I	+0.01365 [+0.01009, +0.01718]
512	10	NeST-WHOLE	DISTS	-0.01082 [-0.01180, -0.00986]
512	19	NeST-WHOLE	PSNR	+0.99391 [+0.93976, +1.04841]
512	19	NeST-WHOLE	LPIPS	-0.03430 [-0.03577, -0.03284]
512	19	NeST-WHOLE	DreamSim	-0.03081 [-0.03328, -0.02843]
512	19	NeST-WHOLE	CLIP-I	+0.02054 [+0.01696, +0.02425]
512	19	NeST-WHOLE	DISTS	-0.01644 [-0.01747, -0.01542]
512	4	NeST-Ada	PSNR	+4.29250 [+4.09816, +4.49208]
512	4	NeST-Ada	LPIPS	-0.19761 [-0.20524, -0.19011]
512	4	NeST-Ada	DreamSim	-0.11068 [-0.12210, -0.09964]
512	4	NeST-Ada	CLIP-I	-0.02007 [-0.02843, -0.01175]
512	4	NeST-Ada	DISTS	-0.06332 [-0.06766, -0.05893]
512	10	NeST-Ada	PSNR	+4.24824 [+4.07079, +4.42766]
512	10	NeST-Ada	LPIPS	-0.14597 [-0.15149, -0.14058]
512	10	NeST-Ada	DreamSim	-0.07443 [-0.08290, -0.06583]
512	10	NeST-Ada	CLIP-I	-0.00466 [-0.01139, +0.00187]
512	10	NeST-Ada	DISTS	-0.04455 [-0.04757, -0.04151]
512	19	NeST-Ada	PSNR	+1.94166 [+1.77815, +2.11000]
512	19	NeST-Ada	LPIPS	-0.03133 [-0.03469, -0.02813]
512	19	NeST-Ada	DreamSim	+0.00723 [+0.00374, +0.01056]
512	19	NeST-Ada	CLIP-I	-0.00339 [-0.00816, +0.00132]
512	19	NeST-Ada	DISTS	+0.00581 [+0.00364, +0.00788]
512	4	WHOLE-Ada	PSNR	+3.78823 [+3.58271, +3.99821]
512	4	WHOLE-Ada	LPIPS	-0.15903 [-0.16820, -0.14991]
512	4	WHOLE-Ada	DreamSim	-0.06007 [-0.07327, -0.04710]
512	4	WHOLE-Ada	CLIP-I	-0.04227 [-0.05108, -0.03354]
512	4	WHOLE-Ada	DISTS	-0.04161 [-0.04673, -0.03637]
512	10	WHOLE-Ada	PSNR	+3.65663 [+3.48147, +3.83501]
512	10	WHOLE-Ada	LPIPS	-0.12290 [-0.12856, -0.11742]
512	10	WHOLE-Ada	DreamSim	-0.05331 [-0.06229, -0.04443]
512	10	WHOLE-Ada	CLIP-I	-0.01831 [-0.02550, -0.01126]
512	10	WHOLE-Ada	DISTS	-0.03374 [-0.03687, -0.03056]
512	19	WHOLE-Ada	PSNR	+0.94774 [+0.77798, +1.12367]
512	19	WHOLE-Ada	LPIPS	+0.00297 [-0.00067, +0.00651]
512	19	WHOLE-Ada	DreamSim	+0.03804 [+0.03381, +0.04212]
512	19	WHOLE-Ada	CLIP-I	-0.02393 [-0.02905, -0.01888]
512	19	WHOLE-Ada	DISTS	+0.02225 [+0.01990, +0.02456]

S2.1 Capacity and rejected receptions

The current N1024 counts retain all 1500 logical events per method/SNR. Valid paid NULL is listed within gray outputs and separately; it is not a lost denominator. Unchanged BPG has 11 body-CRC rejections across the four SNRs, while

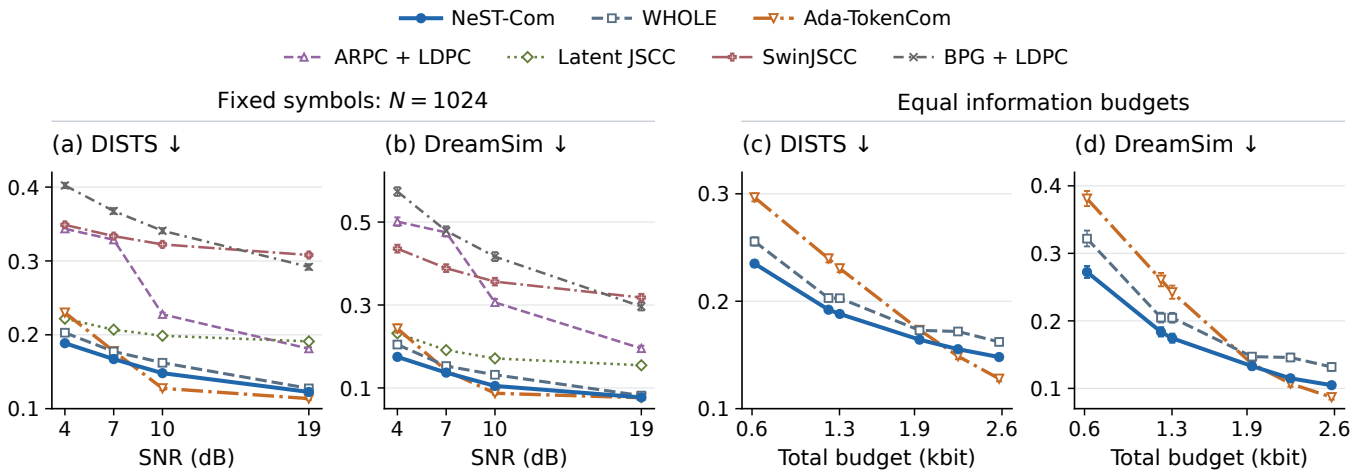


Figure S1: Complete comparison including our digital-link adaptation of the ARPC source codec. The 92 means and pointwise source-bootstrap intervals retain all operating points. ARPC is the closest progressive source-coding precedent; this supplementary curve is its specified digital adaptation.

the other reused external methods have no rejected reconstruction. Complete event counts and native identities are retained in the result files.

Method	SNR	Util. (%)	Gray / 1500	Paid NULL
WHOLE	4	73.812	0	0
WHOLE	7	83.188	11	0
WHOLE	10	77.704	0	0
WHOLE	19	85.240	0	0
NeST	4	99.561	1	0
NeST	7	99.715	7	0
NeST	10	99.794	0	0
NeST	19	99.829	0	0
Ada	4	99.558	0	0
Ada	7	99.670	10	0
Ada	10	98.721	0	0
Ada	19	87.375	0	0

S2.2 Source-level variation

Positive gain is favorable: reference minus NeST for distances and NeST minus reference for PSNR/CLIP-I. Better/tie/worse uses tolerance 10^{-10}, not a perceptual visibility threshold. The distance distributions below include all sources and failures; complete five-metric distributions are in the companion CSV.

SNR	Reference	Metric	Better/tie/worse	Median gain
4	WHOLE	DreamSim	430/1/69	+0.02416
4	WHOLE	DISTS	446/1/53	+0.01356
7	WHOLE	DreamSim	403/3/94	+0.01131
7	WHOLE	DISTS	414/3/83	+0.00795
10	WHOLE	DreamSim	409/16/75	+0.01484
10	WHOLE	DISTS	412/16/72	+0.01130
19	WHOLE	DreamSim	359/6/135	+0.00298
19	WHOLE	DISTS	410/6/84	+0.00434
4	Ada	DreamSim	402/0/98	+0.05383
4	Ada	DISTS	445/0/55	+0.03848
7	Ada	DreamSim	236/0/264	-0.00264

SNR	Reference	Metric	Better/tie/worse	Median gain
7	Ada	DISTS	298/0/202	+0.00696
10	Ada	DreamSim	100/0/400	-0.01489
10	Ada	DISTS	62/0/438	-0.02199
19	Ada	DreamSim	231/0/269	-0.00107
19	Ada	DISTS	144/0/356	-0.00806
4	Latent	DreamSim	427/0/73	+0.05623
4	Latent	DISTS	469/0/31	+0.03244
7	Latent	DreamSim	460/0/40	+0.05402
7	Latent	DISTS	484/0/16	+0.04196
10	Latent	DreamSim	489/0/11	+0.06137
10	Latent	DISTS	500/0/0	+0.05024
19	Latent	DreamSim	500/0/0	+0.07172
19	Latent	DISTS	500/0/0	+0.06790
4	Swin	DreamSim	499/0/1	+0.26441
4	Swin	DISTS	499/0/1	+0.15934
7	Swin	DreamSim	496/0/4	+0.25173
7	Swin	DISTS	493/0/7	+0.17057
10	Swin	DreamSim	500/0/0	+0.24916
10	Swin	DISTS	500/0/0	+0.17445
19	Swin	DreamSim	500/0/0	+0.23573
19	Swin	DISTS	500/0/0	+0.18552
4	BPG	DreamSim	500/0/0	+0.40249
4	BPG	DISTS	499/0/1	+0.21549
7	BPG	DreamSim	499/0/1	+0.34091
7	BPG	DISTS	495/0/5	+0.20147
10	BPG	DreamSim	500/0/0	+0.30439
10	BPG	DISTS	500/0/0	+0.19389
19	BPG	DreamSim	499/0/1	+0.21229
19	BPG	DISTS	500/0/0	+0.17141
4	ARPC	DreamSim	500/0/0	+0.31992
4	ARPC	DISTS	500/0/0	+0.15033
7	ARPC	DreamSim	499/0/1	+0.34055
7	ARPC	DISTS	495/0/5	+0.15890
10	ARPC	DreamSim	499/0/1	+0.18775
10	ARPC	DISTS	498/0/2	+0.08075
19	ARPC	DreamSim	498/0/2	+0.10736
19	ARPC	DISTS	495/0/5	+0.05859

S3 Matched controls and generation sensitivity

S3.1 Endpoint control

The four grids use control100, $N = 1024$, 10 dB, 16-QAM rate 2/3, a common maximum token position $(8, 126)$, and the actual shared noise vectors for repeats 6201–6203. Only legal termination points change. This WHOLE control has five permitted complete-scale endpoints; the main WHOLE system has a different target and must not be substituted. There are 1200 logical outcomes, 891 unique physical inputs, 591 newly executed physical events, and 300 exact physical reuses. Independent RX reconstruction uses 100 new and 197 reused receiver inputs.

Grid	Ends	Util. (%)	Length checks	DISTS	DreamSim
Whole	5	78.133	1.55	0.1650	0.1387
Quarter	20	95.194	4.51	0.1535	0.1143
Eighth	39	97.442	7.57	0.1516	0.1137
Token	352	99.848	125.96	0.1505	0.1115

The quality experiment reuses previously finalized streams, so its new source traversals and independent finalizations are zero. Candidate-set cardinality and descending comparisons must not be described as new network forwards. Fresh deployment computation and measured costs belong to S7. The legacy main-evaluation field named length_checks sometimes records the entire evaluated endpoint set; it is not substituted for the matched-control stopping count shown here.

Comparison	Metric	Difference [95% CI]
Quarter–Whole	PSNR	+0.65169 [0.50944, 0.80674]
Quarter–Whole	LPIPS	-0.02750 [-0.03334, -0.02195]
Quarter–Whole	DreamSim	-0.02447 [-0.03163, -0.01792]
Quarter–Whole	CLIP-I	+0.01678 [0.00815, 0.02602]
Quarter–Whole	DISTS	-0.01153 [-0.01410, -0.00905]
Eighth–Whole	PSNR	+0.75730 [0.59558, 0.93119]
Eighth–Whole	LPIPS	-0.03046 [-0.03672, -0.02443]
Eighth–Whole	DreamSim	-0.02505 [-0.03221, -0.01845]
Eighth–Whole	CLIP-I	+0.01834 [0.00979, 0.02735]
Eighth–Whole	DISTS	-0.01342 [-0.01623, -0.01065]
Token–Whole	PSNR	+0.82337 [0.65877, 1.00101]
Token–Whole	LPIPS	-0.03270 [-0.03923, -0.02649]
Token–Whole	DreamSim	-0.02725 [-0.03419, -0.02071]
Token–Whole	CLIP-I	+0.02339 [0.01484, 0.03271]
Token–Whole	DISTS	-0.01454 [-0.01743, -0.01167]
Eighth–Quarter	PSNR	+0.10561 [0.06774, 0.14656]
Eighth–Quarter	LPIPS	-0.00296 [-0.00408, -0.00194]
Eighth–Quarter	DreamSim	-0.00058 [-0.00201, 0.00082]
Eighth–Quarter	CLIP-I	+0.00156 [-0.00145, 0.00456]
Eighth–Quarter	DISTS	-0.00189 [-0.00279, -0.00106]
Token–Quarter	PSNR	+0.17168 [0.12500, 0.22129]
Token–Quarter	LPIPS	-0.00520 [-0.00653, -0.00390]
Token–Quarter	DreamSim	-0.00278 [-0.00501, -0.00059]
Token–Quarter	CLIP-I	+0.00661 [0.00174, 0.01190]
Token–Quarter	DISTS	-0.00300 [-0.00411, -0.00195]
Token–Eighth	PSNR	+0.06607 [0.03357, 0.10063]
Token–Eighth	LPIPS	-0.00224 [-0.00318, -0.00131]
Token–Eighth	DreamSim	-0.00220 [-0.00422, -0.00014]
Token–Eighth	CLIP-I	+0.00505 [0.00022, 0.01051]
Token–Eighth	DISTS	-0.00112 [-0.00204, -0.00022]

S3.2 Ordered boundary-token interventions

A0 retains prior completion but omits the extra boundary-token correction and updated conditioning. B0 inserts the direct residual correction while keeping the A0 future-condition path. C0 applies both. All three arms use the same actual received prefix and retain failures. This is an ordered intervention on the full received boundary, not an additive decomposition of the tokens beyond Quarter. The complete set contains 1800 logical outcomes. It requires 394 new A0/B0 renders for 197 unique prefixes; two qualification C0 replays are pixel-identical to the official output. No new PHY events are generated.

SNR	Comparison	Metric	Difference [95% CI]
7	B0–A0	PSNR	+0.16706 [0.13852, 0.19873]
7	B0–A0	LPIPS	-0.00639 [-0.00718, -0.00558]
7	B0–A0	DreamSim	-0.00565 [-0.00774, -0.00376]
7	B0–A0	CLIP-I	-0.00061 [-0.00386, 0.00244]
7	B0–A0	DISTS	-0.00294 [-0.00383, -0.00205]
7	C0–B0	PSNR	+0.29197 [0.24095, 0.34673]
7	C0–B0	LPIPS	-0.01079 [-0.01266, -0.00895]
7	C0–B0	DreamSim	-0.00973 [-0.01344, -0.00626]

SNR	Comparison	Metric	Difference [95% CI]
7	C0–B0	CLIP-I	+0.00860 [0.00275, 0.01511]
7	C0–B0	DISTS	-0.00456 [-0.00620, -0.00291]
7	C0–A0	PSNR	+0.45903 [0.39375, 0.52953]
7	C0–A0	LPIPS	-0.01717 [-0.01940, -0.01508]
7	C0–A0	DreamSim	-0.01538 [-0.01982, -0.01135]
7	C0–A0	CLIP-I	+0.00798 [0.00190, 0.01445]
7	C0–A0	DISTS	-0.00750 [-0.00927, -0.00576]
10	B0–A0	PSNR	+0.28681 [0.23964, 0.33531]
10	B0–A0	LPIPS	-0.01225 [-0.01448, -0.01007]
10	B0–A0	DreamSim	-0.01131 [-0.01468, -0.00828]
10	B0–A0	CLIP-I	+0.00190 [-0.00310, 0.00728]
10	B0–A0	DISTS	-0.00621 [-0.00769, -0.00481]
10	C0–B0	PSNR	+0.53657 [0.40447, 0.68156]
10	C0–B0	LPIPS	-0.02045 [-0.02521, -0.01597]
10	C0–B0	DreamSim	-0.01594 [-0.02051, -0.01161]
10	C0–B0	CLIP-I	+0.02149 [0.01389, 0.02963]
10	C0–B0	DISTS	-0.00833 [-0.01022, -0.00647]
10	C0–A0	PSNR	+0.82337 [0.65877, 1.00101]
10	C0–A0	LPIPS	-0.03270 [-0.03923, -0.02649]
10	C0–A0	DreamSim	-0.02725 [-0.03419, -0.02071]
10	C0–A0	CLIP-I	+0.02339 [0.01484, 0.03271]
10	C0–A0	DISTS	-0.01454 [-0.01743, -0.01167]

At $N = 1024$, 19 dB, 1482 of 1500 NeST events have a partial final-scale boundary and 18 transmit the full representation. There is no finer scale to condition, hence B0 and C0 apply the identical final residual map. This structural check introduces no new renders, and historical Quarter A0/B0 means remain labelled Quarter.

S3.3 Ada sampled versus argmax completion

The same 100 source IDs, paid predicted classes, actual received prefixes, three original channel outcomes, models, CFG 4 and numerical route are fixed. Only unreceived-tail selection changes. The greedy context/KV is rebuilt and updated along its own generated path; conditional entropy CDFs do not use CFG. Completed full prefixes and gray outputs are reused. This is a receiver sensitivity control, not additional wireless frames. Table values are argmax minus sampling. At the three tested points, argmax increases both LPIPS and DreamSim. The main comparison retains the original sampling configuration.

N	SNR	Metric	Difference [paired 95% CI]
512	4	PSNR	-0.47494 [-0.80086, -0.15199]
512	4	LPIPS	+0.04232 [+0.02942, +0.05549]
512	4	DreamSim	+0.04502 [+0.02823, +0.06210]
512	4	CLIP-I	-0.02609 [-0.03777, -0.01427]
512	10	PSNR	-0.65111 [-0.98421, -0.33788]
512	10	LPIPS	+0.02955 [+0.02007, +0.03944]
512	10	DreamSim	+0.03454 [+0.02316, +0.04608]
512	10	CLIP-I	-0.02315 [-0.03368, -0.01277]
1024	10	PSNR	-0.15586 [-0.36917, +0.05727]
1024	10	LPIPS	+0.00378 [+0.00067, +0.00707]
1024	10	DreamSim	+0.00358 [+0.00101, +0.00601]
1024	10	CLIP-I	-0.00131 [-0.00523, +0.00253]

S3.4 Matched low-budget termination grid

At N512/10 dB, the same control100 sources, three original noise realizations, fixed 16-QAM rate 2/3, ceiling $(7, 0)$, frozen model and receiver are shared by Whole/Quarter/Eighth/Token. Only allowed termination positions change.

There are 1200 logical positions, 963 unique physical frames (846 exact reuses, 117 new) and 321 unique receiver inputs (282 reuses, 39 new). Thirty-nine new native source/output pairs receive all five metrics. The actual all-candidate length comparisons per image are 4/13/25/126, respectively; these are not model-forward or latency counts.

Grid	Util. (%)	PSNR	LPIPS	DreamSim	CLIP-I	DISTS
Whole	79.176	17.01521	0.29150	0.21809	0.79457	0.20703
Quarter	93.162	17.40397	0.27595	0.20185	0.80128	0.19888
Eighth	96.239	17.47883	0.27253	0.19891	0.80489	0.19739
Token	99.545	17.54887	0.26938	0.19332	0.81030	0.19592

Comparison	Metric	Difference [paired 95% CI]
token-whole	PSNR	+0.53366 [+0.44580, +0.62511]
token-whole	LPIPS	-0.02212 [-0.02553, -0.01874]
token-whole	DreamSim	-0.02477 [-0.03072, -0.01885]
token-whole	CLIP-I	+0.01573 [+0.00749, +0.02374]
token-whole	DISTS	-0.01111 [-0.01339, -0.00887]
eighth-whole	PSNR	+0.46362 [+0.38349, +0.54684]
eighth-whole	LPIPS	-0.01898 [-0.02218, -0.01588]
eighth-whole	DreamSim	-0.01919 [-0.02432, -0.01382]
eighth-whole	CLIP-I	+0.01031 [+0.00234, +0.01795]
eighth-whole	DISTS	-0.00963 [-0.01165, -0.00765]
token-eighth	PSNR	+0.07003 [+0.02910, +0.11490]
token-eighth	LPIPS	-0.00315 [-0.00508, -0.00121]
token-eighth	DreamSim	-0.00558 [-0.00956, -0.00157]
token-eighth	CLIP-I	+0.00541 [-0.00061, +0.01141]
token-eighth	DISTS	-0.00148 [-0.00320, +0.00023]

Eighth retains approximately 87% of the Whole-to-Token mean PSNR gain at this point. Token–Eighth improves PSNR, LPIPS and DreamSim in the paired intervals, while DISTS and CLIP-I intervals cross zero. This diagnostic isolates granularity; it does not decompose the separately calibrated common500 system gain.

S4 Equal information budgets

The information budget includes 12 profile bits, header CRC16, length13 and body CRC16: 57 shared bits, plus 10 class bits for Ada. The arithmetic-code limits are $B - 57$ for WHOLE/NeST and $B - 67$ for Ada. The fixed budgets are $B = 620, 1212, 1302, 1940, 2248, 2577$ bits. WHOLE/NeST share maximum $(10, 0)$; Ada permits 1–256 tokens. Actual information fields and arithmetic streams are independently parsed at RX without PHY noise. The receiver cannot inspect unsent source tokens. This comparison removes wireless and MCS effects, but retains differences among tokenizer, prior, conditioning and decoder.

WHOLE and NeST have one deterministic output per source/budget. Ada uses the three fixed generation anchors 2026103001–2026103003, with the original per-source seed mechanism; metrics, not RGBs, are averaged. There are at most 15 000 logical outputs and 9000 source-level units. These are generation repeats, not new channel realizations.

All 15 000 logical outputs were accepted by the actual source receiver; no NULL or gray output occurs in this error-free diagnostic. New reconstruction requires 195 NeST/WHOLE receiver calls and 8733 Ada calls; 4466 NeST/WHOLE and 225 Ada input identities are reused. There are 8882 newly scored native pairs; 6118 logical positions reuse exact completed metric pairs. These counts are distinct from the number of information-transfer outcomes.

At every budget all five direct NeST–WHOLE intervals favor NeST. NeST–Ada PSNR remains positive throughout. DISTS favors NeST at 620/1212/1302/1940 bits and Ada at 2248/2577 bits. DreamSim favors NeST at 620/1212/1302 bits, crosses zero at 1940 bits, and favors Ada at 2248/2577 bits. LPIPS favors NeST through 2248 bits and crosses zero at 2577 bits. CLIP-I favors Ada at 620/1940/2577 bits, with intervals crossing zero at the other budgets. These mixed results remain part of the complete-system comparison. Increasing the budget from 1212 to 1302 bits raises mean source bits from 915.858 to 918.954 for WHOLE and from 1150.046 to 1239.950 for NeST. Thus NeST uses 89.904 of the extra 90 bits on average, while WHOLE uses 3.096; this is a source mean, not a guarantee for every image.

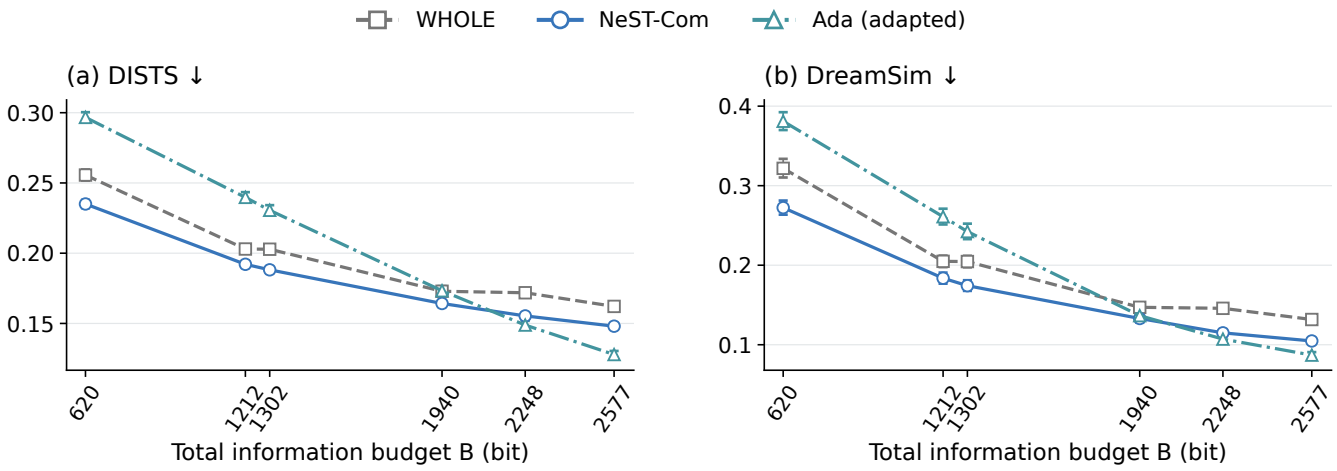


Figure S2: Error-free information-budget comparison on common500. Total bits include all registered information overhead. Numeric budget positions retain their unequal spacing. Points and bars are equal-source means and pointwise 95% source-bootstrap intervals. Ada's three generation repeats are averaged within source; they are not channel repeats. Lower distances are favorable.

S4.1 Complete means and direct paired differences

Bits	Method	Metric	Mean [95% CI]
620	Ada	PSNR	11.34660 [11.16969, 11.53009]
620	Ada	LPIPS	0.54964 [0.54237, 0.55683]
620	Ada	DreamSim	0.38102 [0.36997, 0.39238]
620	Ada	CLIP-I	0.78893 [0.78097, 0.79684]
620	Ada	DISTS	0.29666 [0.29296, 0.30036]
1212	Ada	PSNR	13.06931 [12.86404, 13.27393]
1212	Ada	LPIPS	0.41291 [0.40557, 0.42011]
1212	Ada	DreamSim	0.26099 [0.25122, 0.27104]
1212	Ada	CLIP-I	0.82137 [0.81375, 0.82882]
1212	Ada	DISTS	0.23974 [0.23613, 0.24344]
1302	Ada	PSNR	13.38734 [13.18086, 13.59338]
1302	Ada	LPIPS	0.39382 [0.38649, 0.40102]
1302	Ada	DreamSim	0.24229 [0.23279, 0.25224]
1302	Ada	CLIP-I	0.83018 [0.82262, 0.83736]
1302	Ada	DISTS	0.23054 [0.22700, 0.23419]
1940	Ada	PSNR	16.06878 [15.80180, 16.34242]
1940	Ada	LPIPS	0.26649 [0.25947, 0.27363]
1940	Ada	DreamSim	0.13658 [0.13048, 0.14283]
1940	Ada	CLIP-I	0.86146 [0.85470, 0.86805]
1940	Ada	DISTS	0.17328 [0.17004, 0.17657]
2248	Ada	PSNR	17.74405 [17.45263, 18.03827]
2248	Ada	LPIPS	0.21299 [0.20638, 0.21971]
2248	Ada	DreamSim	0.10710 [0.10246, 0.11197]
2248	Ada	CLIP-I	0.87329 [0.86679, 0.87978]
2248	Ada	DISTS	0.14885 [0.14596, 0.15185]
2577	Ada	PSNR	19.52922 [19.23907, 19.81767]
2577	Ada	LPIPS	0.16729 [0.16181, 0.17293]
2577	Ada	DreamSim	0.08696 [0.08340, 0.09072]
2577	Ada	CLIP-I	0.88382 [0.87767, 0.88999]
2577	Ada	DISTS	0.12787 [0.12544, 0.13041]

Bits	Method	Metric	Mean [95% CI]
620	NeST	PSNR	15.61746 [15.38069, 15.86297]
620	NeST	LPIPS	0.35487 [0.34755, 0.36210]
620	NeST	DreamSim	0.27236 [0.26357, 0.28127]
620	NeST	CLIP-I	0.76969 [0.76034, 0.77914]
620	NeST	DISTS	0.23503 [0.23212, 0.23792]
1212	NeST	PSNR	17.41475 [17.17106, 17.67133]
1212	NeST	LPIPS	0.26496 [0.25874, 0.27117]
1212	NeST	DreamSim	0.18366 [0.17682, 0.19069]
1212	NeST	CLIP-I	0.82110 [0.81230, 0.82951]
1212	NeST	DISTS	0.19210 [0.18954, 0.19466]
1302	NeST	PSNR	17.65585 [17.40777, 17.91721]
1302	NeST	LPIPS	0.25492 [0.24880, 0.26113]
1302	NeST	DreamSim	0.17425 [0.16774, 0.18109]
1302	NeST	CLIP-I	0.82724 [0.81846, 0.83553]
1302	NeST	DISTS	0.18816 [0.18561, 0.19071]
1940	NeST	PSNR	19.04651 [18.79134, 19.31528]
1940	NeST	LPIPS	0.20235 [0.19702, 0.20764]
1940	NeST	DreamSim	0.13301 [0.12797, 0.13834]
1940	NeST	CLIP-I	0.85660 [0.84903, 0.86410]
1940	NeST	DISTS	0.16421 [0.16170, 0.16672]
2248	NeST	PSNR	19.63408 [19.36882, 19.90865]
2248	NeST	LPIPS	0.18322 [0.17818, 0.18827]
2248	NeST	DreamSim	0.11495 [0.11055, 0.11952]
2248	NeST	CLIP-I	0.86890 [0.86184, 0.87572]
2248	NeST	DISTS	0.15536 [0.15290, 0.15784]
2577	NeST	PSNR	20.19536 [19.92833, 20.47116]
2577	NeST	LPIPS	0.16577 [0.16123, 0.17038]
2577	NeST	DreamSim	0.10475 [0.10098, 0.10872]
2577	NeST	CLIP-I	0.87698 [0.87027, 0.88356]
2577	NeST	DISTS	0.14804 [0.14570, 0.15046]
620	WHOLE	PSNR	15.12224 [14.87019, 15.38250]
620	WHOLE	LPIPS	0.39246 [0.38283, 0.40199]
620	WHOLE	DreamSim	0.32180 [0.31030, 0.33372]
620	WHOLE	CLIP-I	0.74806 [0.73814, 0.75801]
620	WHOLE	DISTS	0.25563 [0.25172, 0.25960]
1212	WHOLE	PSNR	16.82131 [16.59007, 17.06388]
1212	WHOLE	LPIPS	0.28808 [0.28216, 0.29399]
1212	WHOLE	DreamSim	0.20486 [0.19764, 0.21207]
1212	WHOLE	CLIP-I	0.80744 [0.79868, 0.81586]
1212	WHOLE	DISTS	0.20297 [0.20052, 0.20552]
1302	WHOLE	PSNR	16.83227 [16.60005, 17.07610]
1302	WHOLE	LPIPS	0.28783 [0.28187, 0.29378]
1302	WHOLE	DreamSim	0.20460 [0.19737, 0.21184]
1302	WHOLE	CLIP-I	0.80770 [0.79894, 0.81618]
1302	WHOLE	DISTS	0.20283 [0.20033, 0.20541]
1940	WHOLE	PSNR	18.55591 [18.31286, 18.81036]
1940	WHOLE	LPIPS	0.21914 [0.21398, 0.22429]
1940	WHOLE	DreamSim	0.14707 [0.14173, 0.15258]
1940	WHOLE	CLIP-I	0.84737 [0.83942, 0.85502]
1940	WHOLE	DISTS	0.17290 [0.17051, 0.17533]
2248	WHOLE	PSNR	18.63777 [18.38451, 18.90215]
2248	WHOLE	LPIPS	0.21754 [0.21220, 0.22296]
2248	WHOLE	DreamSim	0.14580 [0.14038, 0.15132]
2248	WHOLE	CLIP-I	0.84835 [0.84040, 0.85602]
2248	WHOLE	DISTS	0.17184 [0.16935, 0.17438]
2577	WHOLE	PSNR	19.31282 [19.02087, 19.60883]
2577	WHOLE	LPIPS	0.19872 [0.19218, 0.20525]

Bits	Method	Metric	Mean [95% CI]
2577	WHOLE	DreamSim	0.13172 [0.12594, 0.13763]
2577	WHOLE	CLIP-I	0.85774 [0.84973, 0.86557]
2577	WHOLE	DISTS	0.16213 [0.15914, 0.16511]

Bits	Comparison	Metric	Difference [95% CI]
620	NeST–WHOLE	PSNR	+0.49521 [0.43094, 0.56242]
620	NeST–WHOLE	LPIPS	-0.03759 [-0.04141, -0.03380]
620	NeST–WHOLE	DreamSim	-0.04944 [-0.05531, -0.04362]
620	NeST–WHOLE	CLIP-I	+0.02163 [0.01652, 0.02682]
620	NeST–WHOLE	DISTS	-0.02059 [-0.02269, -0.01849]
620	NeST–Ada	PSNR	+4.27086 [4.10364, 4.44710]
620	NeST–Ada	LPIPS	-0.19476 [-0.20160, -0.18812]
620	NeST–Ada	DreamSim	-0.10866 [-0.11918, -0.09816]
620	NeST–Ada	CLIP-I	-0.01924 [-0.02717, -0.01122]
620	NeST–Ada	DISTS	-0.06163 [-0.06509, -0.05818]
620	WHOLE–Ada	PSNR	+3.77565 [3.59676, 3.96826]
620	WHOLE–Ada	LPIPS	-0.15718 [-0.16557, -0.14903]
620	WHOLE–Ada	DreamSim	-0.05922 [-0.07128, -0.04705]
620	WHOLE–Ada	CLIP-I	-0.04087 [-0.04921, -0.03263]
620	WHOLE–Ada	DISTS	-0.04103 [-0.04514, -0.03690]
1212	NeST–WHOLE	PSNR	+0.59345 [0.54574, 0.64438]
1212	NeST–WHOLE	LPIPS	-0.02311 [-0.02455, -0.02166]
1212	NeST–WHOLE	DreamSim	-0.02119 [-0.02371, -0.01875]
1212	NeST–WHOLE	CLIP-I	+0.01366 [0.01008, 0.01720]
1212	NeST–WHOLE	DISTS	-0.01088 [-0.01186, -0.00992]
1212	NeST–Ada	PSNR	+4.34544 [4.17204, 4.52565]
1212	NeST–Ada	LPIPS	-0.14794 [-0.15327, -0.14272]
1212	NeST–Ada	DreamSim	-0.07733 [-0.08526, -0.06928]
1212	NeST–Ada	CLIP-I	-0.00027 [-0.00670, 0.00590]
1212	NeST–Ada	DISTS	-0.04764 [-0.05066, -0.04462]
1212	WHOLE–Ada	PSNR	+3.75200 [3.58339, 3.92664]
1212	WHOLE–Ada	LPIPS	-0.12483 [-0.13038, -0.11947]
1212	WHOLE–Ada	DreamSim	-0.05614 [-0.06466, -0.04776]
1212	WHOLE–Ada	CLIP-I	-0.01392 [-0.02067, -0.00734]
1212	WHOLE–Ada	DISTS	-0.03677 [-0.03992, -0.03365]
1302	NeST–WHOLE	PSNR	+0.82358 [0.76712, 0.88222]
1302	NeST–WHOLE	LPIPS	-0.03292 [-0.03454, -0.03131]
1302	NeST–WHOLE	DreamSim	-0.03035 [-0.03337, -0.02733]
1302	NeST–WHOLE	CLIP-I	+0.01953 [0.01574, 0.02327]
1302	NeST–WHOLE	DISTS	-0.01466 [-0.01575, -0.01359]
1302	NeST–Ada	PSNR	+4.26851 [4.10141, 4.44272]
1302	NeST–Ada	LPIPS	-0.13891 [-0.14393, -0.13391]
1302	NeST–Ada	DreamSim	-0.06804 [-0.07541, -0.06074]
1302	NeST–Ada	CLIP-I	-0.00294 [-0.00857, 0.00249]
1302	NeST–Ada	DISTS	-0.04238 [-0.04529, -0.03948]
1302	WHOLE–Ada	PSNR	+3.44493 [3.28074, 3.61234]
1302	WHOLE–Ada	LPIPS	-0.10599 [-0.11125, -0.10079]
1302	WHOLE–Ada	DreamSim	-0.03769 [-0.04597, -0.02959]
1302	WHOLE–Ada	CLIP-I	-0.02247 [-0.02877, -0.01625]
1302	WHOLE–Ada	DISTS	-0.02772 [-0.03074, -0.02472]
1940	NeST–WHOLE	PSNR	+0.49060 [0.45242, 0.52988]
1940	NeST–WHOLE	LPIPS	-0.01680 [-0.01777, -0.01581]
1940	NeST–WHOLE	DreamSim	-0.01406 [-0.01570, -0.01247]
1940	NeST–WHOLE	CLIP-I	+0.00923 [0.00637, 0.01217]
1940	NeST–WHOLE	DISTS	-0.00870 [-0.00955, -0.00786]

Bits	Comparison	Metric	Difference [95% CI]
1940	NeST–Ada	PSNR	+2.97773 [2.80999, 3.15216]
1940	NeST–Ada	LPIPS	-0.06414 [-0.06812, -0.06032]
1940	NeST–Ada	DreamSim	-0.00357 [-0.00817, 0.00074]
1940	NeST–Ada	CLIP-I	-0.00487 [-0.00963, -0.00018]
1940	NeST–Ada	DISTS	-0.00907 [-0.01158, -0.00668]
1940	WHOLE–Ada	PSNR	+2.48713 [2.31261, 2.66864]
1940	WHOLE–Ada	LPIPS	-0.04735 [-0.05157, -0.04334]
1940	WHOLE–Ada	DreamSim	+0.01049 [0.00563, 0.01511]
1940	WHOLE–Ada	CLIP-I	-0.01409 [-0.01920, -0.00923]
1940	WHOLE–Ada	DISTS	-0.00037 [-0.00304, 0.00222]
2248	NeST–WHOLE	PSNR	+0.99631 [0.94081, 1.05089]
2248	NeST–WHOLE	LPIPS	-0.03432 [-0.03578, -0.03286]
2248	NeST–WHOLE	DreamSim	-0.03085 [-0.03332, -0.02848]
2248	NeST–WHOLE	CLIP-I	+0.02054 [0.01698, 0.02425]
2248	NeST–WHOLE	DISTS	-0.01648 [-0.01751, -0.01546]
2248	NeST–Ada	PSNR	+1.89003 [1.74362, 2.04058]
2248	NeST–Ada	LPIPS	-0.02977 [-0.03302, -0.02669]
2248	NeST–Ada	DreamSim	+0.00784 [0.00454, 0.01100]
2248	NeST–Ada	CLIP-I	-0.00439 [-0.00907, 0.00014]
2248	NeST–Ada	DISTS	+0.00651 [0.00445, 0.00847]
2248	WHOLE–Ada	PSNR	+0.89372 [0.73796, 1.05249]
2248	WHOLE–Ada	LPIPS	+0.00455 [0.00100, 0.00805]
2248	WHOLE–Ada	DreamSim	+0.03869 [0.03458, 0.04270]
2248	WHOLE–Ada	CLIP-I	-0.02493 [-0.02997, -0.02010]
2248	WHOLE–Ada	DISTS	+0.02299 [0.02077, 0.02515]
2577	NeST–WHOLE	PSNR	+0.88254 [0.80859, 0.95733]
2577	NeST–WHOLE	LPIPS	-0.03295 [-0.03563, -0.03024]
2577	NeST–WHOLE	DreamSim	-0.02698 [-0.02991, -0.02409]
2577	NeST–WHOLE	CLIP-I	+0.01924 [0.01555, 0.02311]
2577	NeST–WHOLE	DISTS	-0.01409 [-0.01534, -0.01283]
2577	NeST–Ada	PSNR	+0.66614 [0.55643, 0.78015]
2577	NeST–Ada	LPIPS	-0.00152 [-0.00390, 0.00071]
2577	NeST–Ada	DreamSim	+0.01779 [0.01549, 0.02007]
2577	NeST–Ada	CLIP-I	-0.00685 [-0.01099, -0.00266]
2577	NeST–Ada	DISTS	+0.02017 [0.01851, 0.02177]
2577	WHOLE–Ada	PSNR	-0.21640 [-0.34068, -0.09134]
2577	WHOLE–Ada	LPIPS	+0.03143 [0.02804, 0.03476]
2577	WHOLE–Ada	DreamSim	+0.04477 [0.04100, 0.04860]
2577	WHOLE–Ada	CLIP-I	-0.02608 [-0.03122, -0.02109]
2577	WHOLE–Ada	DISTS	+0.03426 [0.03241, 0.03607]

S4.2 Actual source lengths and endpoint saturation

Bits	Method	Source bits	Padding bits	Full / 500	NULL / 500
620	Ada	547.58	5.42	0	0
1212	Ada	1139.70	5.30	0	0
1302	Ada	1229.54	5.46	0	0
1940	Ada	1866.82	6.18	4	0
2248	Ada	2170.61	10.39	10	0
2577	Ada	2477.89	32.11	75	0
620	NeST	558.28	4.72	0	0
1212	NeST	1150.05	4.95	0	0
1302	NeST	1239.95	5.05	0	0
1940	NeST	1878.36	4.64	0	0

Bits	Method	Source bits	Padding bits	Full / 500	NULL / 500
2248	NeST	2186.54	4.46	0	0
2577	NeST	2515.72	4.28	0	0
620	WHOLE	427.80	135.20	0	0
1212	WHOLE	915.86	239.14	0	0
1302	WHOLE	918.95	326.05	0	0
1940	WHOLE	1566.43	316.57	0	0
2248	WHOLE	1599.71	591.29	0	0
2577	WHOLE	1958.14	561.86	0	0

Full means all 10 NeST/WHOLE scales or all 256 Ada tokens. These counts are computed once per source, not once per Ada repeat. Padding after a complete representation is not failure to fit a prefix. Source bits exclude the separately stated information overhead.

S5 Fixed-policy channel threshold

The 10-dB policies of WHOLE, NeST and Ada remain frozen at actual SNRs 8, 8.5, 9, 9.5 and 10 dB. Each cell has 100 sources and 20 new AWGN vectors, hence 2000 actual complete frames and 30 000 frames in total. The source's original repeat-zero payload, waveform, session, counter and scrambler identity are retained; only noise changes. RX uses the true noise variance. Ada's original repeat-zero generation identity remains fixed across noise and SNR.

All 30 000 real receptions and both distance metrics have completed. There are 17 497 accepted outcomes and 12 503 body/profile-parse rejections, with no header rejection, unexpected accepted source input, source-decoding rejection or NULL. No additional image generation is needed: accepted prefixes and frozen generation identities match existing outputs. Scoring required 88 new native gray-reference pairs for each distance; all other pairs were reused after identity verification.

All methods reject all 2000 receptions at 8 dB. At 8.5 dB their rejection rates are 93.05–94.00%, at 9 dB they are 13.50–15.75%, and at 9.5 dB they are 0.10–0.20%. No rejection is observed at 10 dB. Source-cluster intervals retain all 20 noise repetitions within each sampled source. This finite scan describes the tested fixed policies, not low-error-rate operation or general channel robustness.

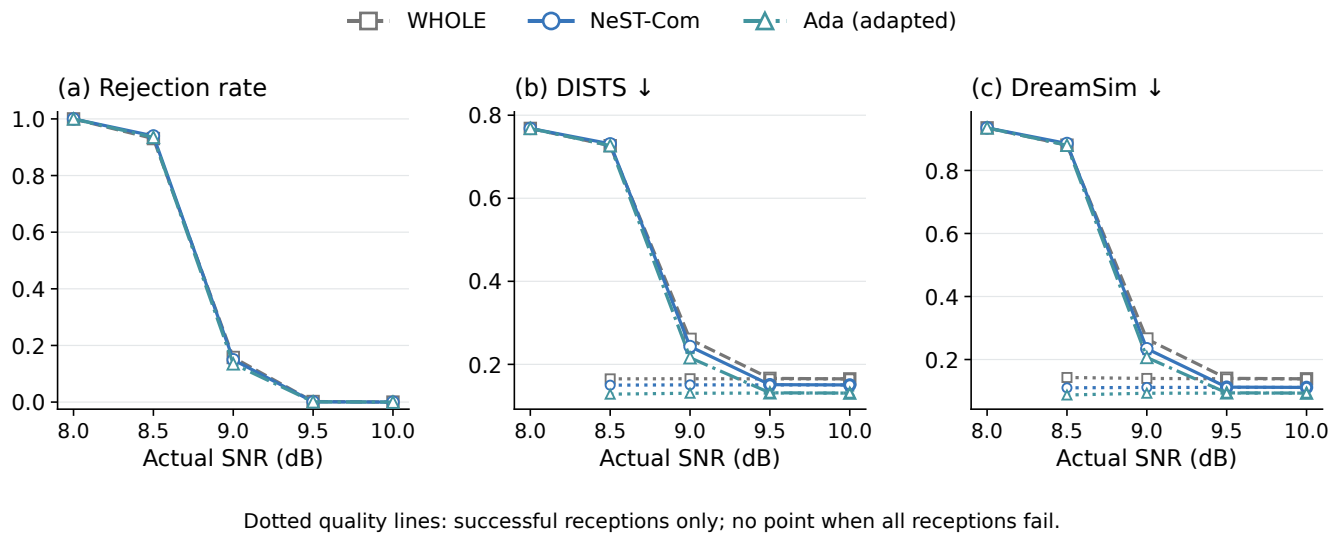


Figure S3: Frozen 10-dB policies evaluated at five actual SNRs. Points and bars retain all receptions and show pointwise 95% source-bootstrap intervals. Dotted quality curves describe successful receptions only and use a changing event denominator; no conditional point exists at 8 dB.

Method	SNR	Metric	Overall mean [95% CI]
Ada	8	Rejection rate	1.00000 [1.00000, 1.00000]
Ada	8	DISTS	0.76824 [0.76057, 0.77563]
Ada	8	DreamSim	0.93515 [0.92703, 0.94316]
Ada	8.5	Rejection rate	0.93500 [0.92200, 0.94700]
Ada	8.5	DISTS	0.72680 [0.71557, 0.73771]
Ada	8.5	DreamSim	0.87996 [0.86721, 0.89279]
Ada	9	Rejection rate	0.13500 [0.11950, 0.15100]
Ada	9	DISTS	0.21603 [0.20456, 0.22794]
Ada	9	DreamSim	0.20653 [0.19141, 0.22233]
Ada	9.5	Rejection rate	0.00100 [0.00000, 0.00250]
Ada	9.5	DISTS	0.13146 [0.12526, 0.13813]
Ada	9.5	DreamSim	0.09424 [0.08616, 0.10328]
Ada	10	Rejection rate	0.00000 [0.00000, 0.00000]
Ada	10	DISTS	0.13079 [0.12463, 0.13742]
Ada	10	DreamSim	0.09341 [0.08534, 0.10239]
NeST	8	Rejection rate	1.00000 [1.00000, 1.00000]
NeST	8	DISTS	0.76824 [0.76057, 0.77563]
NeST	8	DreamSim	0.93515 [0.92703, 0.94316]
NeST	8.5	Rejection rate	0.94000 [0.92950, 0.95000]
NeST	8.5	DISTS	0.73115 [0.72148, 0.74057]
NeST	8.5	DreamSim	0.88560 [0.87443, 0.89682]
NeST	9	Rejection rate	0.14950 [0.13200, 0.16750]
NeST	9	DISTS	0.24294 [0.23094, 0.25502]
NeST	9	DreamSim	0.23428 [0.21806, 0.25085]
NeST	9.5	Rejection rate	0.00100 [0.00000, 0.00250]
NeST	9.5	DISTS	0.15110 [0.14561, 0.15663]
NeST	9.5	DreamSim	0.11231 [0.10429, 0.12089]
NeST	10	Rejection rate	0.00000 [0.00000, 0.00000]
NeST	10	DISTS	0.15047 [0.14502, 0.15601]
NeST	10	DreamSim	0.11149 [0.10345, 0.11997]
WHOLE	8	Rejection rate	1.00000 [1.00000, 1.00000]
WHOLE	8	DISTS	0.76824 [0.76057, 0.77563]
WHOLE	8	DreamSim	0.93515 [0.92703, 0.94316]
WHOLE	8.5	Rejection rate	0.93050 [0.91750, 0.94250]
WHOLE	8.5	DISTS	0.72658 [0.71546, 0.73707]
WHOLE	8.5	DreamSim	0.88020 [0.86695, 0.89269]
WHOLE	9	Rejection rate	0.15750 [0.14300, 0.17200]
WHOLE	9	DISTS	0.26020 [0.25049, 0.27014]
WHOLE	9	DreamSim	0.26497 [0.25173, 0.27887]
WHOLE	9.5	Rejection rate	0.00200 [0.00050, 0.00400]
WHOLE	9.5	DISTS	0.16623 [0.15958, 0.17286]
WHOLE	9.5	DreamSim	0.14037 [0.12846, 0.15272]
WHOLE	10	Rejection rate	0.00000 [0.00000, 0.00000]
WHOLE	10	DISTS	0.16500 [0.15844, 0.17169]
WHOLE	10	DreamSim	0.13874 [0.12689, 0.15117]

Method	SNR	Success / 2000	Cond. DISTS	Cond. DreamSim
Ada	8	0	–	–
Ada	8.5	130	0.12820	0.08738
Ada	9	1730	0.13051	0.09304
Ada	9.5	1998	0.13079	0.09340
Ada	10	2000	0.13079	0.09341
NeST	8	0	–	–
NeST	8.5	120	0.14992	0.11011
NeST	9	1701	0.15061	0.11145

Method	SNR	Success / 2000	Cond. DISTS	Cond. DreamSim
NeST	9.5	1998	0.15047	0.11151
NeST	10	2000	0.15047	0.11149
WHOLE	8	0	–	–
WHOLE	8.5	139	0.16504	0.14297
WHOLE	9	1685	0.16541	0.13993
WHOLE	9.5	1996	0.16503	0.13878
WHOLE	10	2000	0.16500	0.13874

S6 Qualitative examples

Fig.3 uses source135 (C10-016, truck) and source201 (C10-017), chosen from the prior gallery to illustrate favorable NeST perceptual reconstruction. These identities were fixed before unified recalibration and retained afterward. WHOLE/NeST/Ada use the current U10 policies and first registered channel repetition, seed6201; SwinJSCC retains its original first repetition, seed2001. Images show the full 256-by-256 field, with DISTS/DreamSim computed from each displayed native float output. The examples illustrate selected visual gains; common500 means and paired intervals establish population-level comparisons.

Index	Method	PSNR	LPIPS	DreamSim	CLIP-I	DISTS
135	WHOLE	18.16309	0.26054	0.10733	0.87756	0.17963
135	SwinJSCC	21.96407	0.44166	0.36055	0.60724	0.34328
135	Ada-TokenCom	18.81166	0.19548	0.08909	0.83364	0.16033
135	NeST-Com	19.98544	0.21118	0.07420	0.91333	0.14699
201	WHOLE	15.28726	0.26756	0.23034	0.61705	0.17104
201	SwinJSCC	19.74965	0.52086	0.46440	0.60563	0.29948
201	Ada-TokenCom	12.67242	0.26190	0.17663	0.69265	0.15176
201	NeST-Com	16.82297	0.19017	0.17180	0.65959	0.13561

The following supplemental image panels retain their historical policies and their previously fixed selections. Their labels and numbers are not substituted for the newly calibrated main Fig.3. The following panel shows the selected 7-dB television (index 84) and leather (index 143) advantage examples with complete method comparisons. For a complementary view, the next two sources were fixed before their images were opened: common500 N1024/10dB, ranked by each source's mean DISTS gain of Token over WHOLE, with stable source-index tie breaking. We use the package's median-rank source 209 and minimum-gain source 39, without a success filter. Display is the first original repetition (noise 6201 for both methods). Source209 is `n04509417/ILSVRC2012_val_00035733_n04509417`; source 39 is `n02097130/ILSVRC2012_val_00048087_n02097130`. This is a fixed median-rank and DISTS-degradation comparison. Source39 worsens in DISTS but improves in DreamSim, illustrating disagreement between distances.

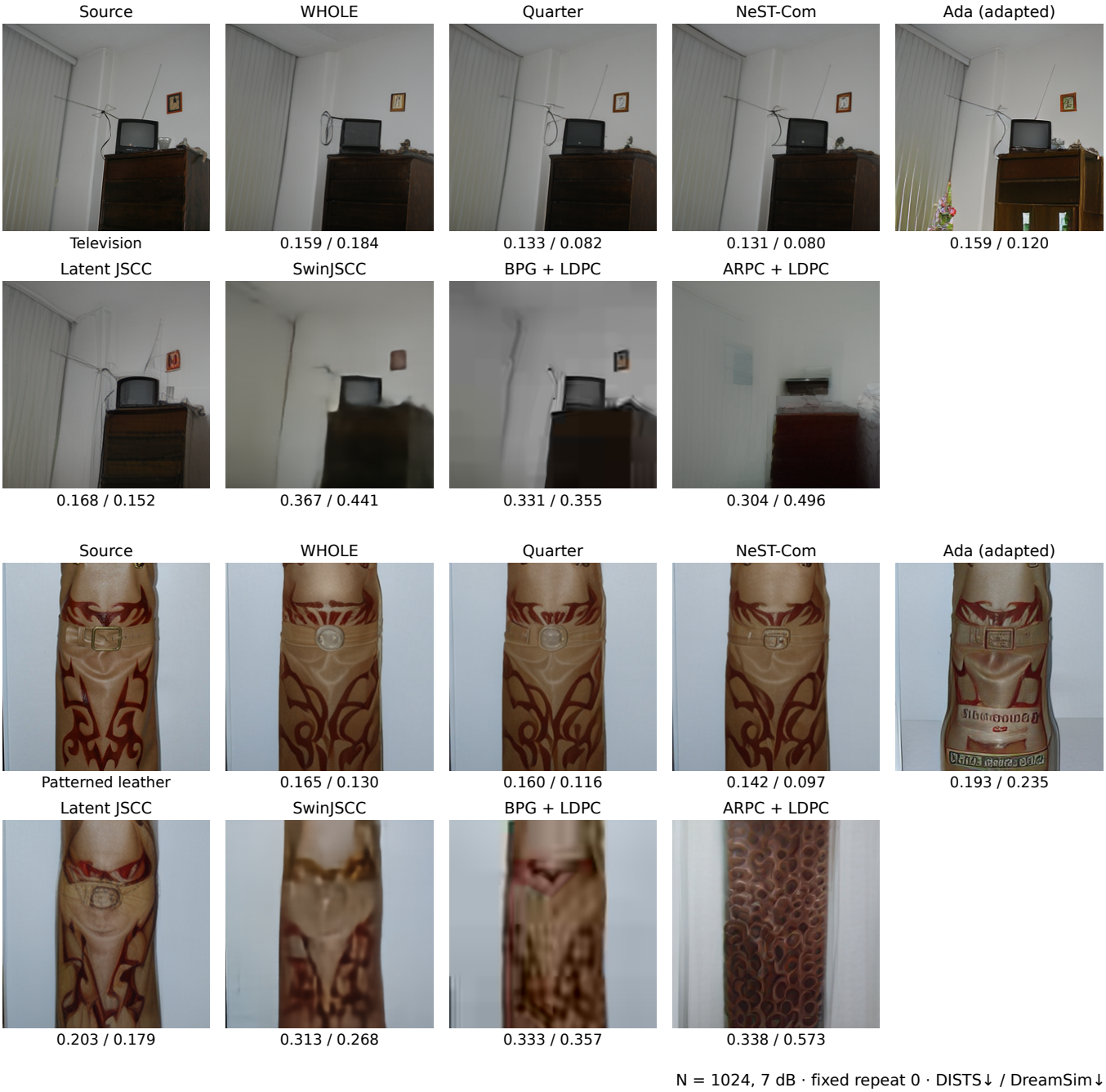


Figure S4: Complete method comparisons for the television and leather advantage examples. Full256×256 fields are retained.



Figure S5: Median-rank (top) and minimum DISTS-gain (bottom) sources at N1024/10dB. Columns are source, WHOLE and NeST. Values are DISTS / DreamSim for the displayed first repetition, lower favorable.

S7 Unified processing costs

The timings below use the H10, H7, L10 and C1024 configurations indexed in S1.3. They measure complete implementations, including candidate generation and scanning. Current N1024 quality uses the separately frozen U policies. Implementation and execution records are archived in the repository reproduction log.

S7.1 Corrected resource protocol for the main comparisons

The resource audit uses the same first 32 common500 IDs at $N = 1024$, 10 dB, batch one, with two warmups and three measurements per source and direction. All five GPU implementations run serially on the same physical RTX PRO 6000. Every method uses CPU affinity [38,40], which selects distinct physical cores 19 and 20 of socket 0, with two intra-op and two inter-op threads. BPG uses these CPU cores without a GPU. The numerical routes, weights, policies, source order and observed RX symbols remain frozen. Every emitted waveform and reconstructed image matches its historical event hash exactly. The qualified compiled Ada implementation is reported separately in S7.4, with its corresponding quality in S7.5.

TX starts with canonical RGB in memory and ends with actual symbols; RX starts with received symbols in memory and ends with RGB. GPU synchronization brackets each span. Tokenization, condition extraction, CDF construction, arithmetic coding, frame fitting, link processing and generation are included. Model loading, input loading, final-output hashing, saved logs and quality scoring are outside the timer. BPG retains source search, codec invocation and RAM serialization. No cached source tokens or reconstructions replace work inside a timed span. TX+RX is the sum of processing spans.

The table gives equal-source means in milliseconds. Brackets for total time are pointwise 95% source-bootstrap intervals (10 000 resamples, fixed seed 2026110201). They describe source variation during this session, not all possible system load. Each listed implementation completes 96 measured RX calls successfully. Full call times, source means, median/p95 and resource telemetry are in `inputs/timing_fairness_20261102/` and the accompanying result directory.

System	TX (ms)	RX (ms)	Total (ms) [95% CI]
WHOLE	215.29	366.42	581.71 [576.77, 586.78]
NeST	248.44	403.99	652.43 [647.45, 657.51]
Latent	15.28	23.93	39.21 [39.20, 39.22]
Swin	21.01	18.16	39.17 [39.07, 39.30]
BPG	2209.97	21.28	2231.26 [2163.61, 2298.68]

S7.2 Matched granularity and audit sensitivity

The matched grid fixes ceiling (8, 126), model, MCS and fitting rule. Whole/Quarter/Eighth TX are measured in the same resource session; Token reuses the identical main NeST TX cell. The main WHOLE policy has a different ceiling and is not substituted for the matched Whole cell. Quality values and all received inputs are unchanged.

Grid	TX (ms)	95% source interval
Whole	185.20	[184.76, 185.67]
Quarter	215.84	[215.33, 216.41]
Eighth	217.76	[217.18, 218.48]
Token	248.44	[246.78, 251.06]

These grid timings include shared scale-wise probability evaluation, independent termination and complete candidate scanning. Earlier serial-Ada and two-source instrumentation diagnostics are retained in `reproduction_logs/timing_execution_history_20261103.md`; they are not used to adjust the 32-source measurements.

S7.3 Prefill compatibility diagnostic

On two fixed sources, conditional-only and original two-branch full-prefix evaluation are compared with the frozen incremental route. Both use identical weights, FP32, math attention and integer-CDF conversion. The reference

contains 256 CDFs and independently terminated streams per source. The table reports exact matches.

Source index	Branches	CDF matches	Stream matches
0	1	0/256	4/256
0	2	0/256	4/256
1	1	0/256	2/256
1	2	0/256	2/256

Full-prefix evaluation changes the CDFs on these inputs. The following study therefore validates a matched prefill transmitter/receiver, including independent recovery of its new bitstreams.

S7.4 Separately qualified compiled Ada implementation

We replace the serial encoder with a single conditional causal prefill and compile the forward operation with `torch.compile`. At RX, the conditional CDF is reconstructed from the received prefix alone using the same tensor shape and strict causal mask; unknown future inputs are public zeros. The generated tail follows the actual recovered context, using the original CFG=4, temperature, filtering and sampling rules. Entropy CDFs use only the conditional branch. We retain FP32, math attention, disabled TF32, the same weights, classifier, tokenizer, image decoder, fixed MCS and candidate set. The compiler may change floating-point execution, so this is a new numerical implementation, not a bit-exact substitution for the original quality-evaluation route.

Qualification recovers all 512 tokens and integer CDFs on two fixed sources and checks both generated tails. The 32-source study uses two warmups and three measurements per direction, with independent RX receiving only its waveform and public configuration. All 160 prefixes, classes and endpoints match the TX reference after independent decoding, and all 96 measured receptions succeed. Repeated outputs are deterministic within this implementation.

Timings include class prediction and digital-link processing at the same synchronized boundaries as S7.1. Model loading and cold compilation are outside these spans. Compilation uses eight CPU cores; timed spans use the same two distinct cores and 2/2 thread pools. S7.5 supplies quality for the corresponding outputs. Intervals use the fixed 10 000-by-32 source-bootstrap bank.

Implementation span	Mean (ms)	95% source interval
TX	114.55	[114.15, 114.93]
RX	2949.76	[2945.89, 2953.91]
TOTAL	3064.30	[3060.46, 3068.37]

The compiled route matches 0/160 historical waveforms and 10/160 reconstructed images exactly; S7.5 pairs its latency with its own output quality and explains the seed dependence. These are implementation-specific processing measurements. Ada's published 19/1012-ms TX/RX values use different hardware and processing boundaries.

S7.5 Quality paired with compiled Ada timing

We restore the 32 saved timed received waveforms and compare native outputs on the exact same sources, received prefixes/classes and channel events. The restored compiled outputs match all 32 timed output hashes. No TX, new channel event or new timing run is performed. The original and compiled coders recover identical source tokens, but their bitstreams differ; the existing seed derivation from received payload bytes can therefore change the tail-generation seed.

A fixed-original-seed control isolates payload-derived seed changes. It produces byte-identical native images on all 32 sources and zero differences on all five metrics. The actual timed outputs differ on 30 sources; their paired mean differences below all have intervals crossing zero. Each source contributes one distinct quality event. This diagnostic pairs quality with the compiled implementation's measured cost; the common500 results retain the original implementation.

Arm difference	Metric	Mean [paired 95% CI]
Compiled-Original	PSNR	-0.03839 [-0.47062, +0.39219]

Arm difference	Metric	Mean [paired 95% CI]
Compiled-Original	LPIPS	-0.00168 [-0.00524, +0.00177]
Compiled-Original	DreamSim	-0.00349 [-0.00837, +0.00082]
Compiled-Original	CLIP-I	-0.00103 [-0.00614, +0.00379]
Compiled-Original	DISTS	-0.00175 [-0.00444, +0.00090]
Fixed-seed-Original	PSNR	+0.00000 [+0.00000, +0.00000]
Fixed-seed-Original	LPIPS	+0.00000 [+0.00000, +0.00000]
Fixed-seed-Original	DreamSim	+0.00000 [+0.00000, +0.00000]
Fixed-seed-Original	CLIP-I	+0.00000 [+0.00000, +0.00000]
Fixed-seed-Original	DISTS	+0.00000 [+0.00000, +0.00000]
Compiled-Fixed-seed	PSNR	-0.03839 [-0.47062, +0.39219]
Compiled-Fixed-seed	LPIPS	-0.00168 [-0.00524, +0.00177]
Compiled-Fixed-seed	DreamSim	-0.00349 [-0.00837, +0.00082]
Compiled-Fixed-seed	CLIP-I	-0.00103 [-0.00614, +0.00379]
Compiled-Fixed-seed	DISTS	-0.00175 [-0.00444, +0.00090]

The actual compiled timed outputs have source-mean PSNR 19.36069, LPIPS 0.16355, DreamSim 0.09640, CLIP-I 0.87857, DISTS 0.13098. Their corresponding TX/RX/total means are 114.55/2949.76/3064.30 ms. The fixed-seed diagnostic is not assigned a newly measured latency.

S7.6 Historical E5 resource protocol

The historical 7-dB, N512, ARPC and grid-RX groups below were not rerun. For completeness, the original N1024/10 groups are also retained and labeled historical. They use affinity [38,39] (two SMT siblings of one physical core), six intra-op and two inter-op threads. These values are not combined with the corrected-resource main tables.

Timing uses the first 32 common500 IDs, batch one, two warmups and three measurements per source, one actual RTX PRO 6000 and two SMT siblings of one physical CPU core. Original numerical routes are retained; six library compute threads are confined to those two logical CPUs. Model loading, disk I/O, logging and quality scoring are excluded. TX starts with in-memory canonical RGB and ends with actual transmitted symbols. RX starts with in-memory received symbols and ends with canonical RGB. GPU boundary synchronization surrounds each measured span; TX+RX is a processing sum, not radio propagation latency.

The measured operations include tokenizer, condition extraction, CDF/arithmetic coding, finalization/fitting, paid packaging, LDPC/QAM and real receiver PHY, independent source decoding, completion and image decoding. Ada includes ConvNeXt prediction; ARPC includes caption generation; BPG includes its actual search and codec invocation. BPG is CPU-only on the same host and CPU quota; necessary codec serialization uses RAM files. Fresh timing does not substitute cached source tokens or reconstructions. Successful-RX conditional time is reported separately from fast rejection. Means first average within source; median/p95 use measured calls with linear quantiles.

All 16 groups contribute 3072 measured boundary spans after 2048 warmups. ARPC includes its original model runtime and CPU-PHY interprocess transfer within the same SMT-pair affinity. BPG includes codec invocation and source search.

S7.7 Historical measured groups

Values below are milliseconds. Means average the three measurements within each of 32 sources; median and p95 are calculated from the 96 measured calls. Total pairs the TX and RX spans for the same source and repetition. The success count refers to measured RX calls; conditional RX means exclude rejected receptions and are not substituted for overall RX means.

System	N	SNR	Span	Mean (ms)	Median	p95
WHOLE	1024	7	TX	263.35	262.01	270.94
WHOLE	1024	7	RX	375.26	374.77	379.50
WHOLE	1024	7	total	638.61	637.56	646.96
NeST	1024	7	TX	362.09	360.53	370.70

System	N	SNR	Span	Mean (ms)	Median	p95
NeST	1024	7	RX	390.86	390.64	394.25
NeST	1024	7	total	752.95	751.30	761.83
Ada	1024	7	TX	7546.57	7537.00	7627.76
Ada	1024	7	RX	7873.85	7864.66	7932.79
Ada	1024	7	total	15420.41	15404.73	15537.01
WHOLE	1024	10	TX	213.83	213.37	216.53
WHOLE	1024	10	RX	369.90	371.15	387.29
WHOLE	1024	10	total	583.73	585.23	602.20
NeST	1024	10	TX	248.38	247.09	251.45
NeST	1024	10	RX	411.64	401.40	431.95
NeST	1024	10	total	660.02	649.47	678.57
Ada	1024	10	TX	7795.61	7790.73	7854.45
Ada	1024	10	RX	7626.11	7623.30	7667.14
Ada	1024	10	total	15421.72	15412.42	15498.35
WHOLE	512	10	TX	157.33	157.00	159.30
WHOLE	512	10	RX	325.48	325.10	328.35
WHOLE	512	10	total	482.81	482.32	485.94
NeST	512	10	TX	163.34	163.14	164.79
NeST	512	10	RX	348.89	348.50	352.27
NeST	512	10	total	512.22	511.81	515.45
Ada	512	10	TX	7378.20	7373.40	7423.35
Ada	512	10	RX	7789.73	7785.50	7836.21
Ada	512	10	total	15167.93	15159.94	15239.95
ARPC	1024	10	TX	661.72	653.02	732.59
ARPC	1024	10	RX	797.03	796.61	803.32
ARPC	1024	10	total	1458.76	1448.62	1531.35
Swin	1024	10	TX	20.53	20.47	20.67
Swin	1024	10	RX	17.61	17.60	17.73
Swin	1024	10	total	38.14	38.08	38.41
Latent	1024	10	TX	15.42	15.41	15.47
Latent	1024	10	RX	24.10	24.05	24.21
Latent	1024	10	total	39.51	39.47	39.64
BPG	1024	10	TX	2441.98	2457.79	2827.31
BPG	1024	10	RX	31.71	31.58	33.13
BPG	1024	10	total	2473.69	2489.51	2859.12
Grid Whole	1024	10	TX	183.50	182.95	186.63
Grid Whole	1024	10	RX	378.16	380.12	393.96
Grid Whole	1024	10	total	561.66	569.55	578.15
Grid Quarter	1024	10	TX	214.09	213.20	216.88
Grid Quarter	1024	10	RX	385.21	383.68	388.92
Grid Quarter	1024	10	total	599.30	597.40	609.10
Grid Eighth	1024	10	TX	217.25	216.96	219.00
Grid Eighth	1024	10	RX	388.82	386.04	414.90
Grid Eighth	1024	10	total	606.07	603.35	630.91

System	N	SNR	Success / 96	Conditional RX (ms)
WHOLE	1024	7	96	375.26
NeST	1024	7	96	390.86
Ada	1024	7	96	7873.85
WHOLE	1024	10	96	369.90
NeST	1024	10	96	411.64
Ada	1024	10	96	7626.11
WHOLE	512	10	96	325.48
NeST	512	10	96	348.89
Ada	512	10	96	7789.73

System	N	SNR	Success / 96	Conditional RX (ms)
ARPC	1024	10	96	797.03
Swin	1024	10	96	17.61
Latent	1024	10	96	24.10
BPG	1024	10	96	31.71
Grid Whole	1024	10	96	378.16
Grid Quarter	1024	10	96	385.21
Grid Eighth	1024	10	96	388.82

S7.8 Shared computation accounting

The matched grid comparison uses ceiling $(8, 126)$; its token arm reuses the identical NeST timing group at $N = 1024$, 10 dB. The main WHOLE timing group has its own family target and is not the matched-grid WHOLE control. The timed implementation constructs all candidate streams and scans the complete list for feasibility before selecting the longest one. Its recorded legacy field `length_checks` is the selected endpoint's descending rank, not the actual number of tests. The corrected accounting records 5/20/39/352 candidate tests for matched Whole/Quarter/Eighth/Token, separately from the early-stopping comparisons used by the cached quality control in S3. Both rules select the same endpoint; the timing includes this implementation's full scan.

Fresh source-probability traversals, CDF construction steps, independent finalizations, actual tests and endpoint ranks are in `source_fitting_accounting.csv`, derived from the bound code and per-trial witnesses. Stage-level instrumented model/PHY calls are in `model_call_accounting.csv`; they are not exhaustive operator counts. Counters include warmups and are not interchangeable with latency. No prior source-output cache replaces timed model work.

This software implementation includes actual CDF construction, independent arithmetic termination and real digital decoding. The result is a deployment-cost comparison on this host and these original runtimes, not an optimized-hardware throughput bound. The table reports the observed success counts and descriptive timing statistics.

S8 Transfer to Kodak24

Kodak retains the earlier ImageNet-calibrated policies, without repeating the new N1024 calibration on Kodak. It uses the original RGB center 256×256 crop without resizing before cropping. ImageNet-calibrated policies, model components and receiver configuration remain fixed. WHOLE and Quarter are the prior complete-scale and quarter-grid systems, respectively; Token uses integer endpoints. Ada remains the paid predicted-class fixed-frame adaptation. SNR 4/10/19 each use three original events per image (seeds2001/2002/2003), giving 216 logical results per method. Native outputs and existing four metrics are reused under exact identities; DISTS is evaluated on all 269 unique source/output pairs. Token needs 213 new physical frames plus 3 exact reused frames, and 71 new unique formal reconstructions plus 1 reused reconstruction. All 864 four-method events are accepted; the independent source population is 24, not 72.

Token improves PSNR, DISTS and DreamSim over WHOLE at all three SNRs. Compared with Quarter, Token's DISTS/DreamSim intervals exclude zero at 10/19 dB and cross zero at 4 dB. Against Ada, Token improves DISTS/DreamSim at 4 dB; Ada improves DISTS at 10/19 dB, while DreamSim differences cross zero. The complete five-metric results follow; all intervals use the existing 10 000×24 source bank after source-level event averaging.

SNR	Method	Metric	Mean [pointwise 95% CI]
4	WHOLE	PSNR	18.22252 [17.19135, 19.28486]
4	WHOLE	DISTS	0.20677 [0.19716, 0.21593]
4	WHOLE	LPIPS	0.29033 [0.26444, 0.31649]
4	WHOLE	DreamSim	0.20424 [0.17572, 0.23450]
4	WHOLE	CLIP-I	0.81941 [0.79144, 0.84729]
10	WHOLE	PSNR	20.33540 [19.00247, 21.72279]
10	WHOLE	DISTS	0.16732 [0.15447, 0.17974]
10	WHOLE	LPIPS	0.20294 [0.17231, 0.23357]
10	WHOLE	DreamSim	0.12658 [0.10627, 0.14680]
10	WHOLE	CLIP-I	0.85987 [0.83716, 0.88284]
19	WHOLE	PSNR	22.64749 [21.42126, 23.92284]

SNR	Method	Metric	Mean [pointwise 95% CI]
19	WHOLE	DISTS	0.12954 [0.12145, 0.13784]
19	WHOLE	LPIPS	0.12688 [0.11148, 0.14242]
19	WHOLE	DreamSim	0.08077 [0.06859, 0.09424]
19	WHOLE	CLIP-I	0.89736 [0.87628, 0.91739]
4	Quarter	PSNR	18.61812 [17.58014, 19.68850]
4	Quarter	DISTS	0.19469 [0.18526, 0.20428]
4	Quarter	LPIPS	0.27138 [0.24587, 0.29674]
4	Quarter	DreamSim	0.17664 [0.15190, 0.20216]
4	Quarter	CLIP-I	0.83837 [0.81319, 0.86334]
10	Quarter	PSNR	21.00895 [19.79903, 22.26482]
10	Quarter	DISTS	0.15675 [0.14639, 0.16737]
10	Quarter	LPIPS	0.17623 [0.15438, 0.19836]
10	Quarter	DreamSim	0.11016 [0.09240, 0.12951]
10	Quarter	CLIP-I	0.86917 [0.84631, 0.89104]
19	Quarter	PSNR	22.87538 [21.59650, 24.19178]
19	Quarter	DISTS	0.12711 [0.11842, 0.13604]
19	Quarter	LPIPS	0.12304 [0.10683, 0.13951]
19	Quarter	DreamSim	0.07949 [0.06724, 0.09289]
19	Quarter	CLIP-I	0.89545 [0.87353, 0.91613]
4	Token	PSNR	18.78991 [17.77901, 19.83027]
4	Token	DISTS	0.19566 [0.18560, 0.20525]
4	Token	LPIPS	0.26054 [0.23491, 0.28639]
4	Token	DreamSim	0.17165 [0.14835, 0.19668]
4	Token	CLIP-I	0.84106 [0.81457, 0.86665]
10	Token	PSNR	21.12961 [19.93020, 22.36030]
10	Token	DISTS	0.15356 [0.14400, 0.16359]
10	Token	LPIPS	0.17156 [0.14997, 0.19328]
10	Token	DreamSim	0.10473 [0.08826, 0.12314]
10	Token	CLIP-I	0.88441 [0.86585, 0.90309]
19	Token	PSNR	23.05219 [21.77905, 24.34964]
19	Token	DISTS	0.12508 [0.11634, 0.13390]
19	Token	LPIPS	0.11866 [0.10295, 0.13449]
19	Token	DreamSim	0.07746 [0.06447, 0.09191]
19	Token	CLIP-I	0.89827 [0.87404, 0.92010]
4	Ada	PSNR	13.53319 [12.61366, 14.57350]
4	Ada	DISTS	0.25729 [0.24236, 0.27212]
4	Ada	LPIPS	0.45443 [0.42323, 0.48381]
4	Ada	DreamSim	0.30625 [0.26196, 0.35335]
4	Ada	CLIP-I	0.78734 [0.75053, 0.82311]
10	Ada	PSNR	19.92384 [18.48343, 21.42034]
10	Ada	DISTS	0.13142 [0.11836, 0.14511]
10	Ada	LPIPS	0.17394 [0.14562, 0.20325]
10	Ada	DreamSim	0.10152 [0.08147, 0.12556]
10	Ada	CLIP-I	0.88105 [0.85052, 0.90936]
19	Ada	PSNR	21.46358 [20.29288, 22.67017]
19	Ada	DISTS	0.11340 [0.10648, 0.12014]
19	Ada	LPIPS	0.14084 [0.12365, 0.15827]
19	Ada	DreamSim	0.08271 [0.07083, 0.09517]
19	Ada	CLIP-I	0.89815 [0.87737, 0.91827]

SNR	Comparison	Metric	Difference [paired 95% CI]
4	Token minus WHOLE	PSNR	+0.56739 [+0.39543, +0.74525]
4	Token minus WHOLE	DISTS	-0.01111 [-0.01626, -0.00593]
4	Token minus WHOLE	LPIPS	-0.02979 [-0.03607, -0.02319]
4	Token minus WHOLE	DreamSim	-0.03259 [-0.04669, -0.01907]

SNR	Comparison	Metric	Difference [paired 95% CI]
4	Token minus WHOLE	CLIP-I	+0.02165 [+0.01043, +0.03289]
10	Token minus WHOLE	PSNR	+0.79421 [+0.50913, +1.09128]
10	Token minus WHOLE	DISTS	-0.01376 [-0.01986, -0.00788]
10	Token minus WHOLE	LPIPS	-0.03138 [-0.04223, -0.02047]
10	Token minus WHOLE	DreamSim	-0.02185 [-0.03313, -0.01098]
10	Token minus WHOLE	CLIP-I	+0.02454 [+0.01321, +0.03776]
19	Token minus WHOLE	PSNR	+0.40470 [+0.30712, +0.50909]
19	Token minus WHOLE	DISTS	-0.00447 [-0.00648, -0.00260]
19	Token minus WHOLE	LPIPS	-0.00822 [-0.01067, -0.00601]
19	Token minus WHOLE	DreamSim	-0.00331 [-0.00549, -0.00113]
19	Token minus WHOLE	CLIP-I	+0.00091 [-0.00623, +0.00705]
4	Token minus Quarter	PSNR	+0.17179 [+0.05088, +0.31324]
4	Token minus Quarter	DISTS	+0.00097 [-0.00384, +0.00561]
4	Token minus Quarter	LPIPS	-0.01085 [-0.01598, -0.00597]
4	Token minus Quarter	DreamSim	-0.00499 [-0.01620, +0.00392]
4	Token minus Quarter	CLIP-I	+0.00269 [-0.01127, +0.01503]
10	Token minus Quarter	PSNR	+0.12065 [+0.07247, +0.17387]
10	Token minus Quarter	DISTS	-0.00320 [-0.00624, -0.00044]
10	Token minus Quarter	LPIPS	-0.00467 [-0.00801, -0.00151]
10	Token minus Quarter	DreamSim	-0.00542 [-0.00960, -0.00162]
10	Token minus Quarter	CLIP-I	+0.01524 [+0.00610, +0.02431]
19	Token minus Quarter	PSNR	+0.17680 [+0.12165, +0.23538]
19	Token minus Quarter	DISTS	-0.00204 [-0.00369, -0.00062]
19	Token minus Quarter	LPIPS	-0.00438 [-0.00637, -0.00265]
19	Token minus Quarter	DreamSim	-0.00203 [-0.00364, -0.00032]
19	Token minus Quarter	CLIP-I	+0.00282 [-0.00206, +0.00751]
4	Token minus Ada	PSNR	+5.25672 [+4.60102, +5.93444]
4	Token minus Ada	DISTS	-0.06163 [-0.07486, -0.04864]
4	Token minus Ada	LPIPS	-0.19389 [-0.21446, -0.17418]
4	Token minus Ada	DreamSim	-0.13460 [-0.17473, -0.09885]
4	Token minus Ada	CLIP-I	+0.05372 [+0.02390, +0.08339]
10	Token minus Ada	PSNR	+1.20577 [+0.66352, +1.84723]
10	Token minus Ada	DISTS	+0.02214 [+0.01196, +0.03122]
10	Token minus Ada	LPIPS	-0.00238 [-0.01419, +0.00860]
10	Token minus Ada	DreamSim	+0.00321 [-0.01324, +0.01611]
10	Token minus Ada	CLIP-I	+0.00336 [-0.01758, +0.02570]
19	Token minus Ada	PSNR	+1.58860 [+1.39589, +1.78090]
19	Token minus Ada	DISTS	+0.01168 [+0.00768, +0.01569]
19	Token minus Ada	LPIPS	-0.02218 [-0.02645, -0.01836]
19	Token minus Ada	DreamSim	-0.00524 [-0.01259, +0.00326]
19	Token minus Ada	CLIP-I	+0.00012 [-0.01260, +0.01418]
4	Quarter minus WHOLE	PSNR	+0.39560 [+0.23628, +0.54342]
4	Quarter minus WHOLE	DISTS	-0.01208 [-0.01674, -0.00753]
4	Quarter minus WHOLE	LPIPS	-0.01894 [-0.02466, -0.01299]
4	Quarter minus WHOLE	DreamSim	-0.02760 [-0.03997, -0.01587]
4	Quarter minus WHOLE	CLIP-I	+0.01896 [+0.00878, +0.02989]
10	Quarter minus WHOLE	PSNR	+0.67356 [+0.40750, +0.95752]
10	Quarter minus WHOLE	DISTS	-0.01057 [-0.01587, -0.00576]
10	Quarter minus WHOLE	LPIPS	-0.02671 [-0.03701, -0.01667]
10	Quarter minus WHOLE	DreamSim	-0.01642 [-0.02618, -0.00721]
10	Quarter minus WHOLE	CLIP-I	+0.00930 [-0.00355, +0.02510]
19	Quarter minus WHOLE	PSNR	+0.22790 [+0.14025, +0.32419]
19	Quarter minus WHOLE	DISTS	-0.00243 [-0.00391, -0.00110]
19	Quarter minus WHOLE	LPIPS	-0.00383 [-0.00563, -0.00223]
19	Quarter minus WHOLE	DreamSim	-0.00127 [-0.00294, +0.00034]
19	Quarter minus WHOLE	CLIP-I	-0.00190 [-0.00760, +0.00222]
4	Quarter minus Ada	PSNR	+5.08493 [+4.42344, +5.78332]

SNR	Comparison	Metric	Difference [paired 95% CI]
4	Quarter minus Ada	DISTS	-0.06259 [-0.07638, -0.04927]
4	Quarter minus Ada	LPIPS	-0.18304 [-0.20325, -0.16354]
4	Quarter minus Ada	DreamSim	-0.12961 [-0.16912, -0.09369]
4	Quarter minus Ada	CLIP-I	+0.05103 [+0.02591, +0.07808]
10	Quarter minus Ada	PSNR	+1.08511 [+0.56422, +1.69685]
10	Quarter minus Ada	DISTS	+0.02533 [+0.01622, +0.03347]
10	Quarter minus Ada	LPIPS	+0.00229 [-0.00922, +0.01301]
10	Quarter minus Ada	DreamSim	+0.00864 [-0.00883, +0.02165]
10	Quarter minus Ada	CLIP-I	-0.01188 [-0.02758, +0.00382]
19	Quarter minus Ada	PSNR	+1.41180 [+1.22595, +1.60036]
19	Quarter minus Ada	DISTS	+0.01371 [+0.00969, +0.01776]
19	Quarter minus Ada	LPIPS	-0.01779 [-0.02225, -0.01377]
19	Quarter minus Ada	DreamSim	-0.00321 [-0.01010, +0.00507]
19	Quarter minus Ada	CLIP-I	-0.00269 [-0.01363, +0.00935]
4	WHOLE minus Ada	PSNR	+4.68933 [+4.03011, +5.37820]
4	WHOLE minus Ada	DISTS	-0.05052 [-0.06303, -0.03884]
4	WHOLE minus Ada	LPIPS	-0.16410 [-0.18559, -0.14359]
4	WHOLE minus Ada	DreamSim	-0.10201 [-0.14076, -0.06682]
4	WHOLE minus Ada	CLIP-I	+0.03207 [+0.00520, +0.05914]
10	WHOLE minus Ada	PSNR	+0.41155 [-0.15764, +1.06101]
10	WHOLE minus Ada	DISTS	+0.03590 [+0.02542, +0.04549]
10	WHOLE minus Ada	LPIPS	+0.02900 [+0.01505, +0.04273]
10	WHOLE minus Ada	DreamSim	+0.02506 [+0.00693, +0.04098]
10	WHOLE minus Ada	CLIP-I	-0.02118 [-0.04404, +0.00135]
19	WHOLE minus Ada	PSNR	+1.18390 [+1.02522, +1.34532]
19	WHOLE minus Ada	DISTS	+0.01615 [+0.01219, +0.01997]
19	WHOLE minus Ada	LPIPS	-0.01396 [-0.01862, -0.00960]
19	WHOLE minus Ada	DreamSim	-0.00194 [-0.00915, +0.00644]
19	WHOLE minus Ada	CLIP-I	-0.00079 [-0.01303, +0.01192]

SNR	Comparison	Metric	Better/tie/worse	Median gain
4	Token vs WHOLE	PSNR	23/0/1	+0.58584
4	Token vs WHOLE	DISTS	18/0/6	+0.01016
4	Token vs WHOLE	LPIPS	23/0/1	+0.03394
4	Token vs WHOLE	DreamSim	21/0/3	+0.02514
4	Token vs WHOLE	CLIP-I	20/0/4	+0.01863
10	Token vs WHOLE	PSNR	22/0/2	+0.77979
10	Token vs WHOLE	DISTS	18/0/6	+0.01544
10	Token vs WHOLE	LPIPS	21/0/3	+0.04019
10	Token vs WHOLE	DreamSim	19/0/5	+0.01697
10	Token vs WHOLE	CLIP-I	21/0/3	+0.01717
19	Token vs WHOLE	PSNR	24/0/0	+0.44240
19	Token vs WHOLE	DISTS	20/0/4	+0.00378
19	Token vs WHOLE	LPIPS	23/0/1	+0.00659
19	Token vs WHOLE	DreamSim	18/0/6	+0.00273
19	Token vs WHOLE	CLIP-I	16/0/8	+0.00226
4	Token vs Quarter	PSNR	14/0/10	+0.06004
4	Token vs Quarter	DISTS	13/0/11	+0.00089
4	Token vs Quarter	LPIPS	21/0/3	+0.00921
4	Token vs Quarter	DreamSim	12/0/12	+0.00097
4	Token vs Quarter	CLIP-I	15/0/9	+0.00541
10	Token vs Quarter	PSNR	21/0/3	+0.08544
10	Token vs Quarter	DISTS	15/0/9	+0.00078
10	Token vs Quarter	LPIPS	18/0/6	+0.00318
10	Token vs Quarter	DreamSim	15/0/9	+0.00301

SNR	Comparison	Metric	Better/tie/worse	Median gain
10	Token vs Quarter	CLIP-I	19/0/5	+0.01124
19	Token vs Quarter	PSNR	22/1/1	+0.14001
19	Token vs Quarter	DISTS	16/1/7	+0.00162
19	Token vs Quarter	LPIPS	21/1/2	+0.00274
19	Token vs Quarter	DreamSim	17/1/6	+0.00226
19	Token vs Quarter	CLIP-I	13/1/10	+0.00175
4	Token vs Ada	PSNR	24/0/0	+5.19071
4	Token vs Ada	DISTS	23/0/1	+0.05022
4	Token vs Ada	LPIPS	24/0/0	+0.19064
4	Token vs Ada	DreamSim	23/0/1	+0.10509
4	Token vs Ada	CLIP-I	19/0/5	+0.04465
10	Token vs Ada	PSNR	17/0/7	+1.01499
10	Token vs Ada	DISTS	3/0/21	-0.02454
10	Token vs Ada	LPIPS	13/0/11	+0.00181
10	Token vs Ada	DreamSim	7/0/17	-0.00732
10	Token vs Ada	CLIP-I	13/0/11	+0.00184
19	Token vs Ada	PSNR	24/0/0	+1.55558
19	Token vs Ada	DISTS	3/0/21	-0.01190
19	Token vs Ada	LPIPS	24/0/0	+0.02016
19	Token vs Ada	DreamSim	15/0/9	+0.00406
19	Token vs Ada	CLIP-I	8/0/16	-0.00586